

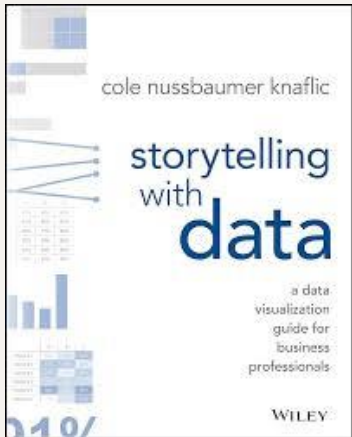
Data Visualization and Storytelling

Professor Laura Ruiz
MSc Business Analytics



March 24, 2026

SESSION 5: How to pick the right chart, how visualizations work





How the Values Compare to Each Other



How Values Relate to Each Other

looker

What are you trying to show?

How the Data is Distributed



How the Data is Composed



How to pick the right chart?

The visualization (or viz) you create depends on:

- **Your data type:** Categorical, numerical, time-based, or geographic
- **Your analysis goal:** Compare values, show trends, reveal relationships, or track progress
- **Your audience:** Executives need high-level summaries, while analysts need detailed data

Main chart categories

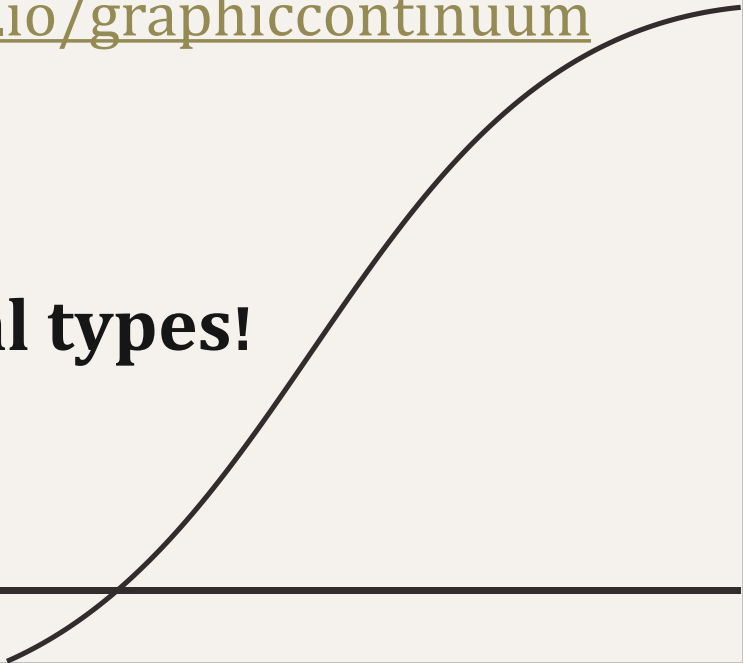
- Comparisons
- Trends
- Proportions
- Relationships
- Distribution
- Deviation
- Geographic

These categories
are broad and
there is
crossover
between them!

Useful references for chart
selection guides:

- <http://bhv.io/datavizproject>
- <http://bhv.io/vizvocabulary>
- <http://bhv.io/graphiccontinuum>

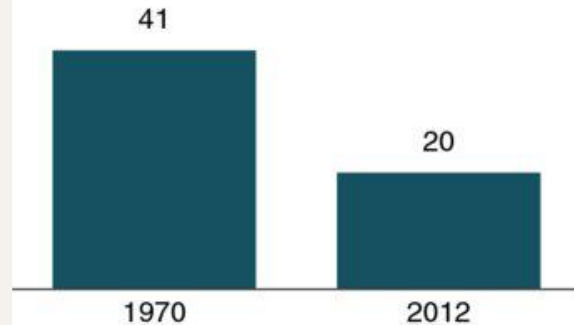
Power BI offers more than 40 built-in visual types!



Simple text or numbers

Children with a "Traditional" Stay-at- Home Mother

% of children with a married
stay-at-home mother with a
working husband



Note: Based on children younger than 18.
Their mothers are categorized based on
employment status in 1970 and 2012.

Source: Pew Research Center analysis of
March Current Population Surveys
Integrated Public Use Microdata Series
(IPUMS-CPS), 1971 and 2013

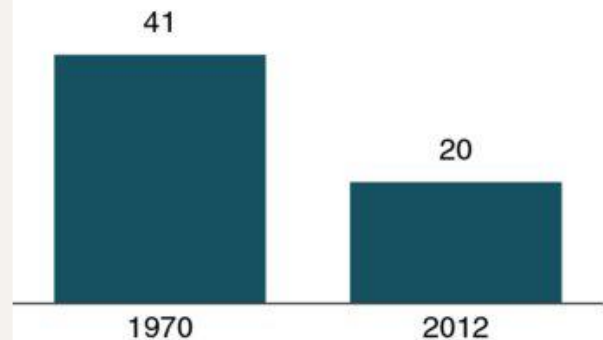
Adapted from PEW RESEARCH CENTER

When you have just a number or two that you want to communicate: *use the numbers directly.*
The fact that you have some numbers does not mean that you need a graph!

Simple text or numbers

Children with a "Traditional" Stay-at-Home Mother

% of children with a married stay-at-home mother with a working husband



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from PEW RESEARCH CENTER

20%

of children had a
traditional stay-at-home mom
in 2012, compared to 41% in 1970

Simple text or numbers

Key metrics and progress

When to use: You need to highlight a single important value or track progress toward a goal.

Best visuals:

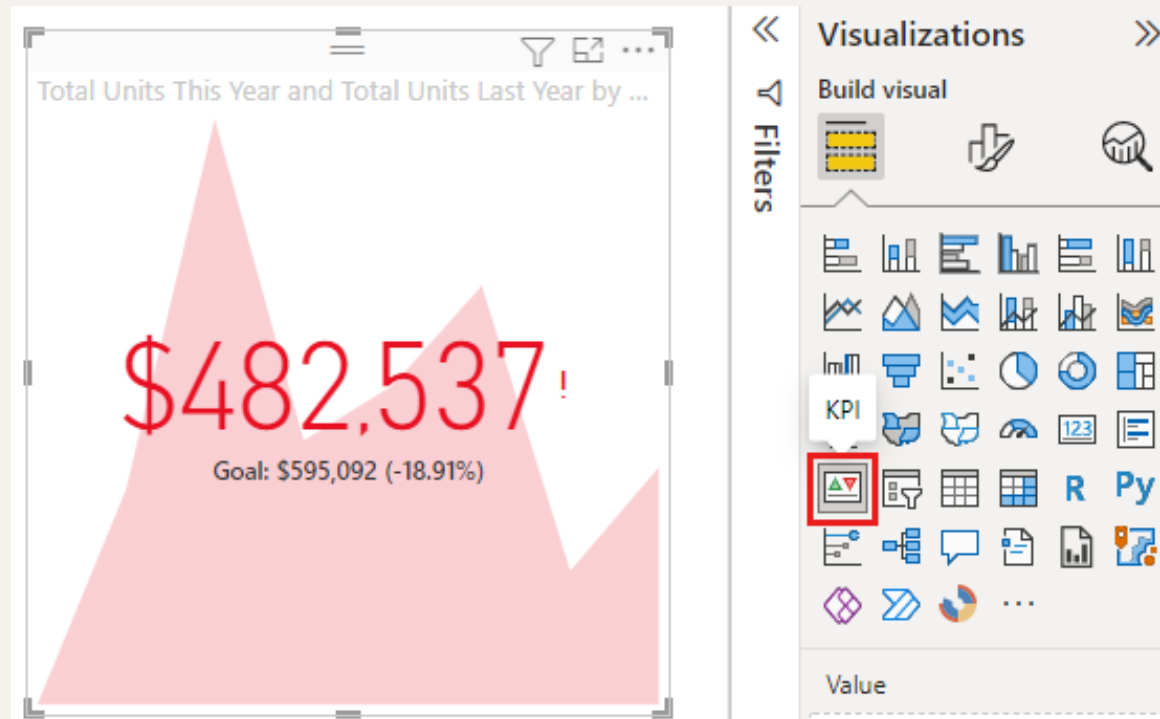
- **Card visuals**: Display a single important number prominently. Use for key metrics that don't need context.
- **KPI visuals**: Show current value, target, and status. Ideal for tracking performance against goals.
- **Gauge charts**: Display a single value's progress toward a goal using a radial gauge.

Example scenarios:

- Total revenue or profit
 - Customer satisfaction score
 - Sales quota attainment
 - Project completion percentage
-

Simple text or numbers

Key metrics and progress: where to find in Power BI



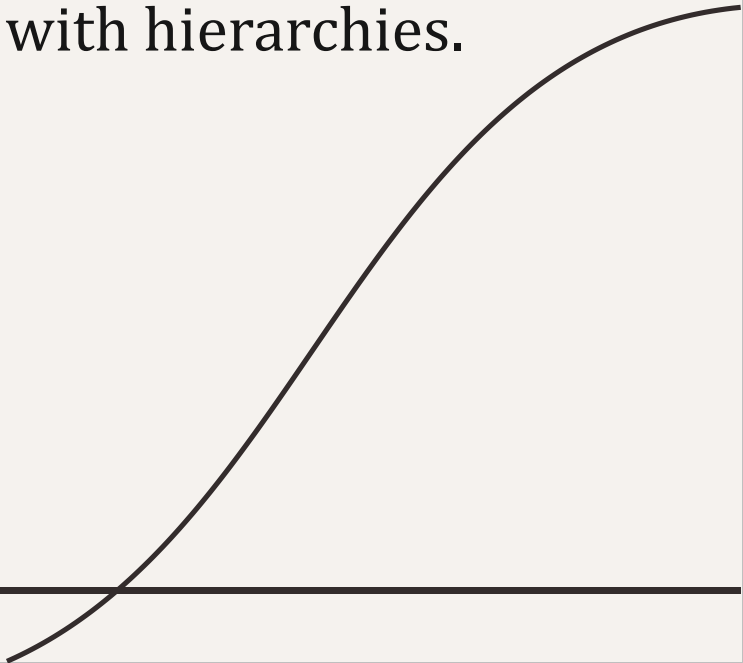
Detailed data and exact values

When to use: Your audience needs to see exact numbers, multiple measures, or drill into detailed records.

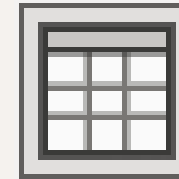
Best visuals:

- **Tables**: Display detailed data in rows and columns. Use when exact values are more important than visual patterns.
- **Matrix visuals**: Show data across multiple dimensions with hierarchies. Automatically aggregates and supports drill-down.

Example scenarios:

- Product catalog with multiple attributes
 - Financial statements with subtotals
 - Employee roster with demographics
 - Transaction details
- 
-

Detailed data and exact values: Tables



- You want the design to fade into the background, letting the data take center stage.
- Don't let heavy borders or shading compete for attention. Instead, use light borders or simply white space to set apart elements of the table.

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Light borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Minimal borders

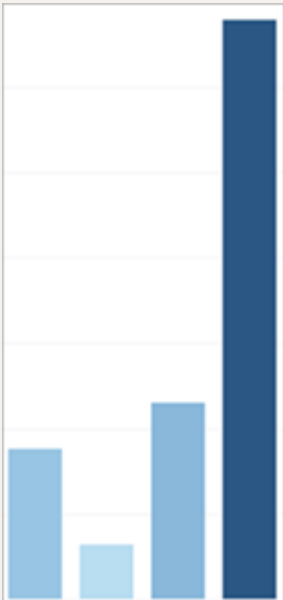
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Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Tables in live presentations

Using a table in a live presentation is rarely a good idea. As your audience reads it, you lose their ears and attention to make your point verbally. When you find yourself using a table in a presentation or report, ask yourself: what is the point you are trying to make? Odds are that there will be a better way to pull out and visualize the piece or pieces of interest. In the event that you feel you're losing too much by doing this, consider whether including the full table in the appendix and a link or reference to it will meet your audience's needs.

Comparisons & rankings

- Most frequent purpose for visualizations is to make comparisons
- Categorical data ranked in ascending or descending order.
- **Magnitude:** shows the relative size or value of two or more discrete items. For example, if you are comparing sales for different regions, you are looking at magnitude.

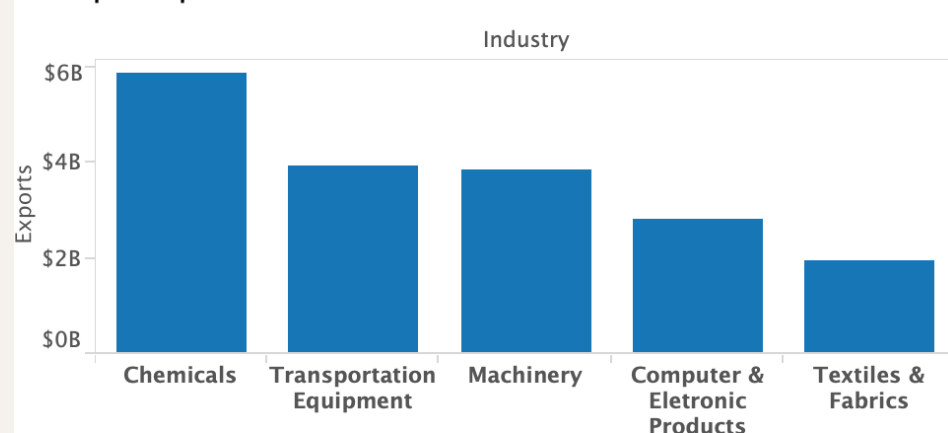


- Which of these dimension members has the highest measure?
- Are there any exceptional dimensions?
- How large of a gap is there between the lowest and highest measure between these dimensions?

Comparisons & rankings

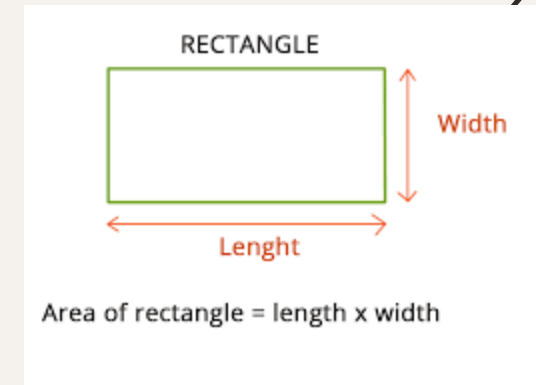
- Most frequent purpose for visualizations is to make comparisons.
- **Ranking:** Sometimes you not only want to depict the magnitude of some value, but also the relative ranking of all the members of your dimension.
- **Example:** Showing the top five export industries

NC top 5 export industries



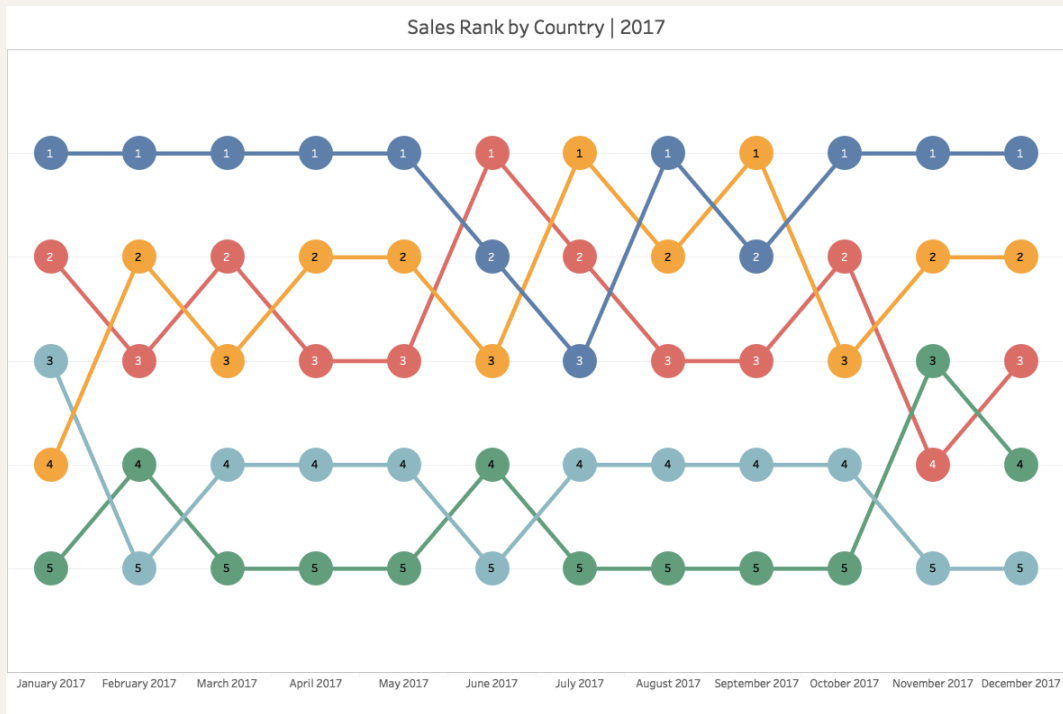
Bar chart or column chart: WHY SO UBIQUITOUS?

We are good at measuring the area of rectangles (brain science) –
We are terrible with circles



Comparisons & rankings

- Most frequent purpose for visualizations is to make comparisons.
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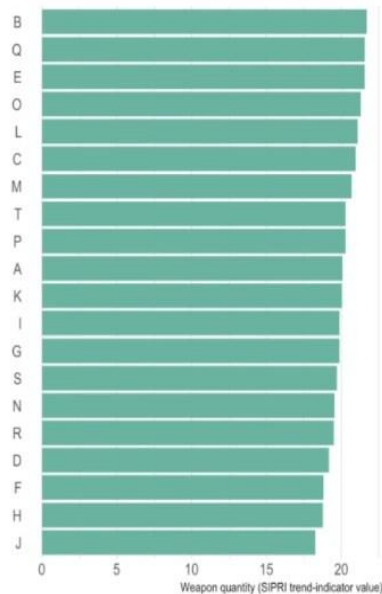
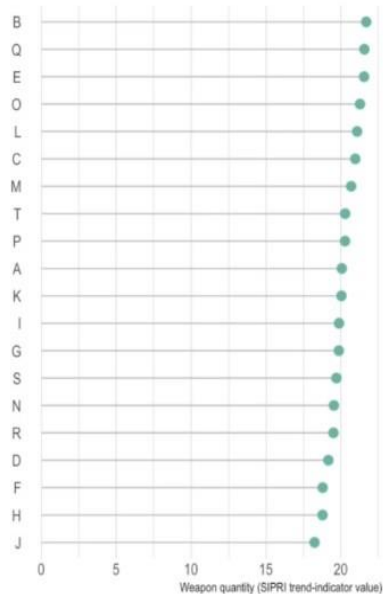
Bumpchart

https://www.youtube.com/watch?v=iDFyLbhD_IM

Comparisons & rankings: Beyond bar charts

Barplot > Variation: the lollipop plot

The bar is replaced with a line ending with a dot.



Good idea because

- ✓ Good for dense barplots with similar values
- ✓ Principle of minimum ink
- ✓ It is clearer that we do not have to decode area
- ✓ Decode position in a common scale

#1



#3



Lollipop chart



Co

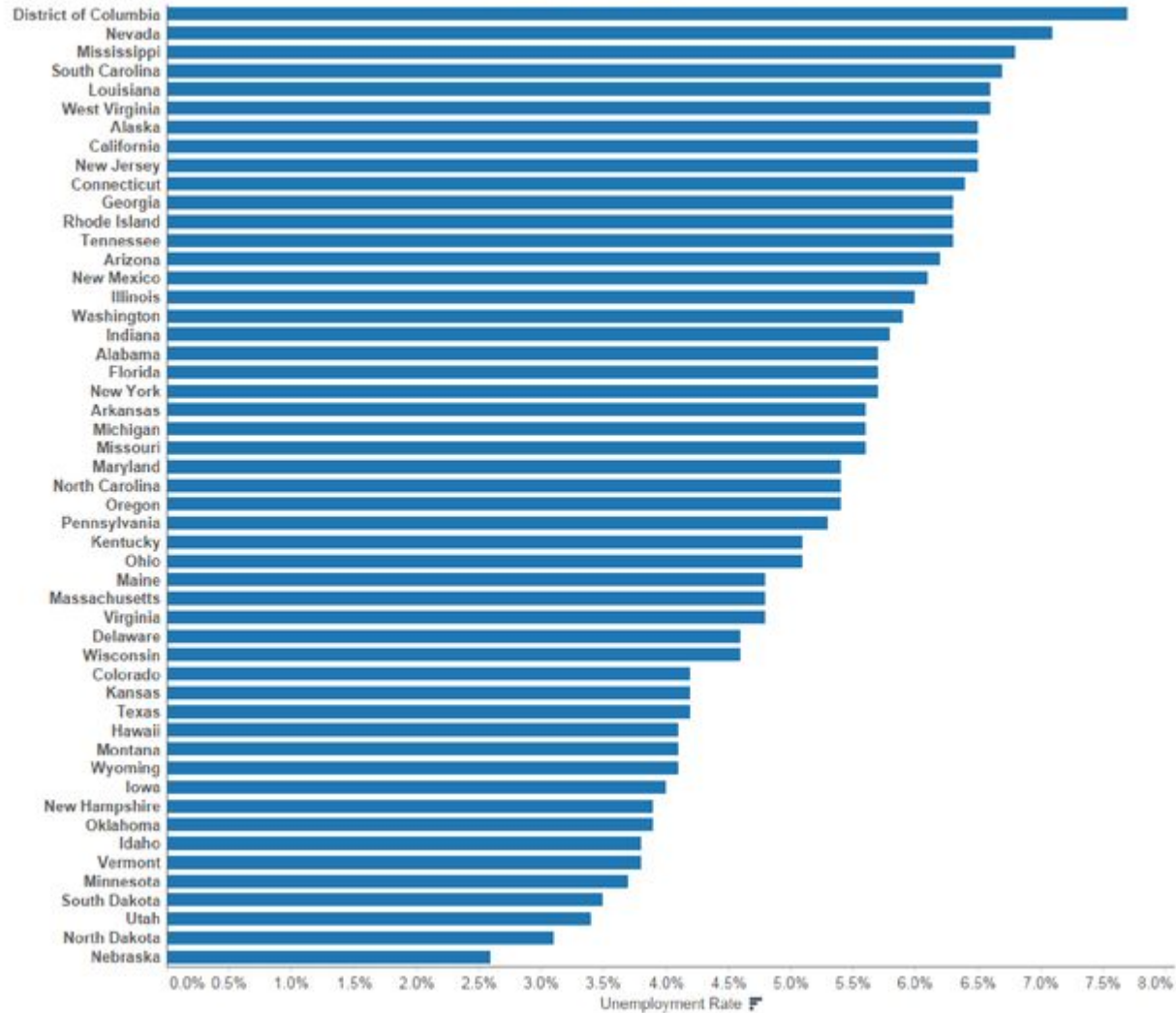


Bar

The bar is replac



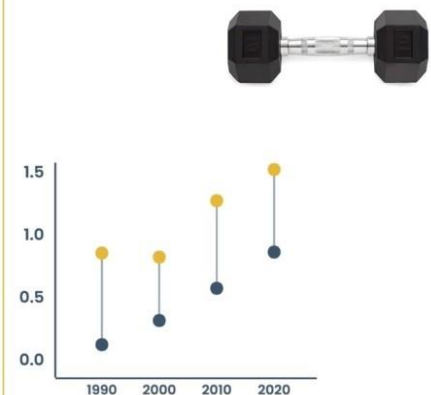
Unemployment by State



Comparisons & rankings: Beyond bar charts

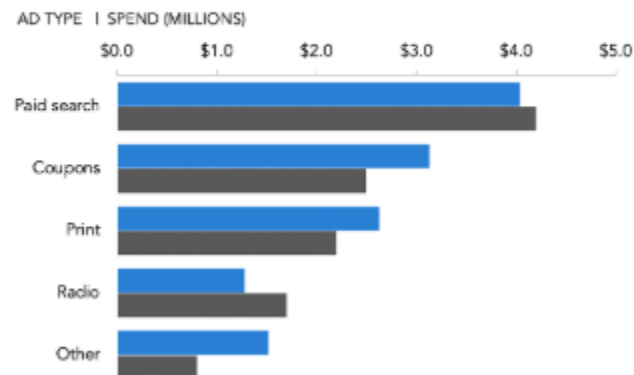
Dot plot

The Cleveland dot plot – or – dumbbell



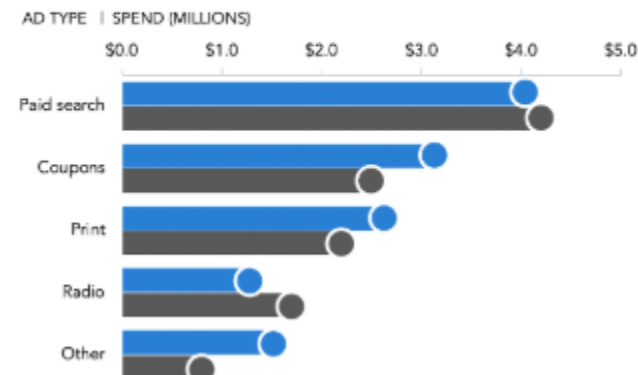
A dual-series bar chart...

Advertising spend by category: us vs. competitor



Encode end points with dots.

Advertising spend by category: us vs. competitor



Collapse, remove bars, connect.

Advertising spend by category: us vs. competitor

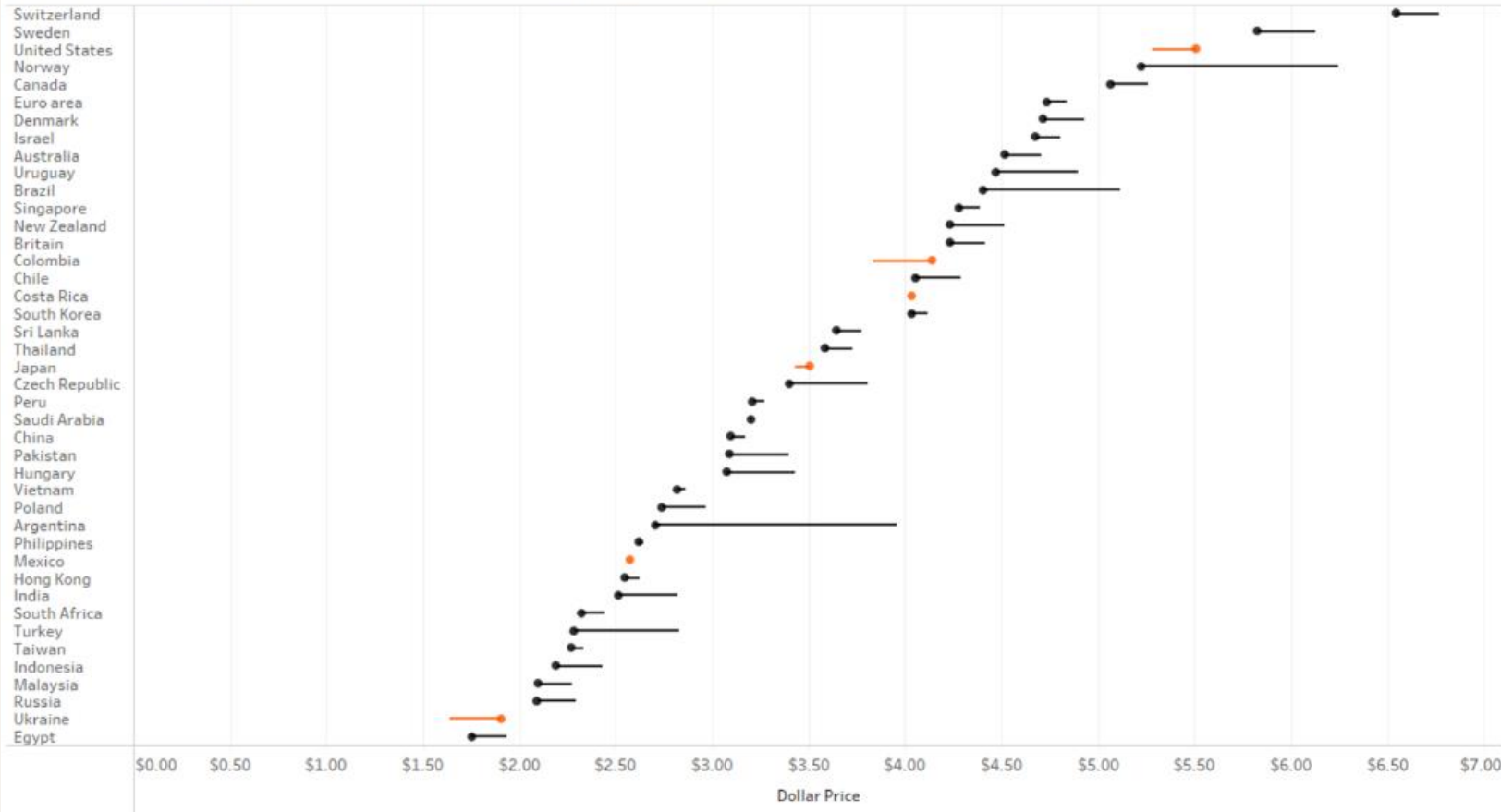


The connected dot plot encodes the absolute values (via positions of the dots), but puts emphasis on the ***difference or change*** between the data series due to the visual connection between the dots.

Comparisons & rankings: Beyond bar charts

Where has the dollar price of the Big Mac **increased** between Jan 2018 and July 2018?

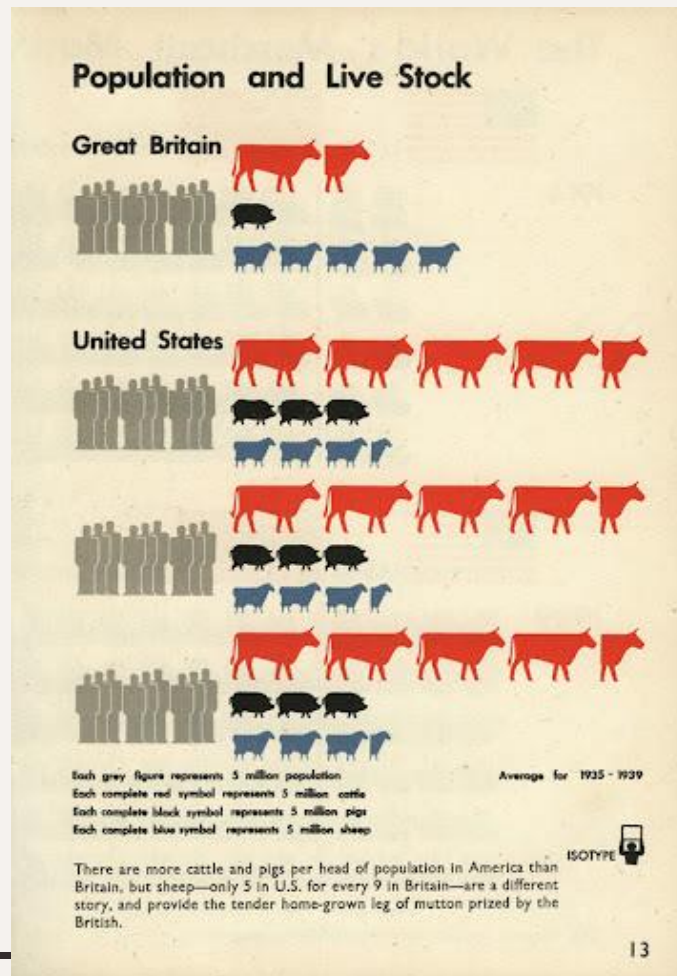
Dot represents the July 2018 position, line starts at the Jan 2018 position



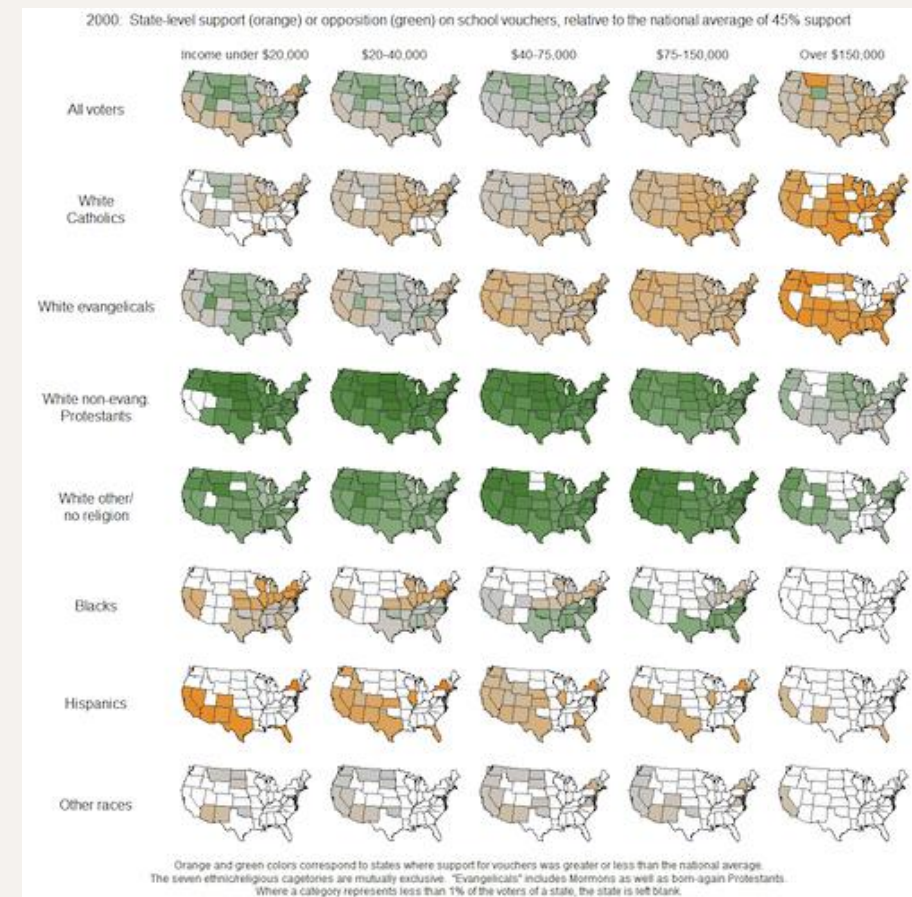
Data Source: McDonald's; The Economist
Project: #MakeoverMonday & #SWDChallenge
Design: @shortenl

Comparisons & rankings: Beyond bar charts

Isotypes



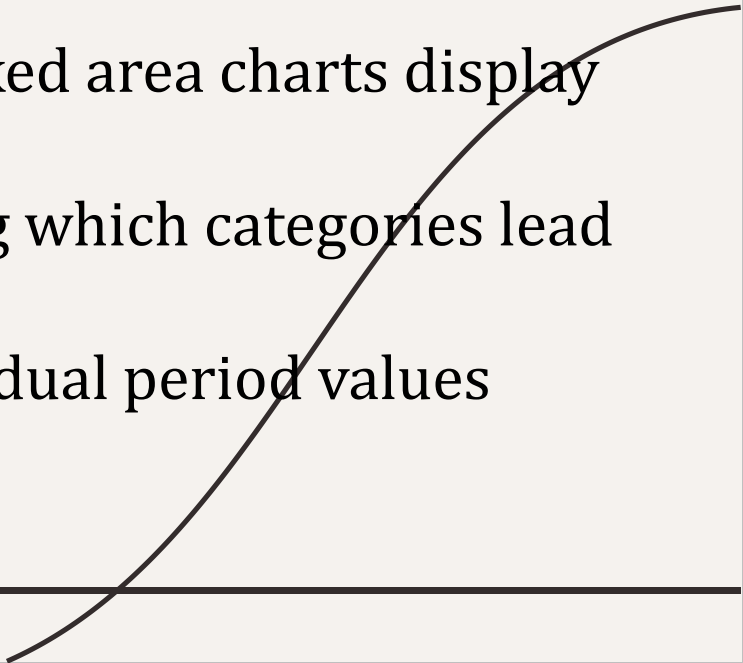
Small Multiples



Time-based trends

- Showing a change over time for a measure is one of the fundamental categories of visualizations.

Best visuals:

- **Line charts**: Emphasize the overall shape and pattern of change. Ideal for continuous time series data.
 - **Area charts**: Show magnitude of change over time. Stacked area charts display multiple series contributing to a total.
 - **Ribbon charts**: Display rank changes over time, showing which categories lead in each period.
 - **Column charts**: Use when you want to emphasize individual period values rather than continuous trends.
- 

Trends: Line chart

Sometimes the originality doesn't reside in the way you present data but rather in what you discover in the visualization process



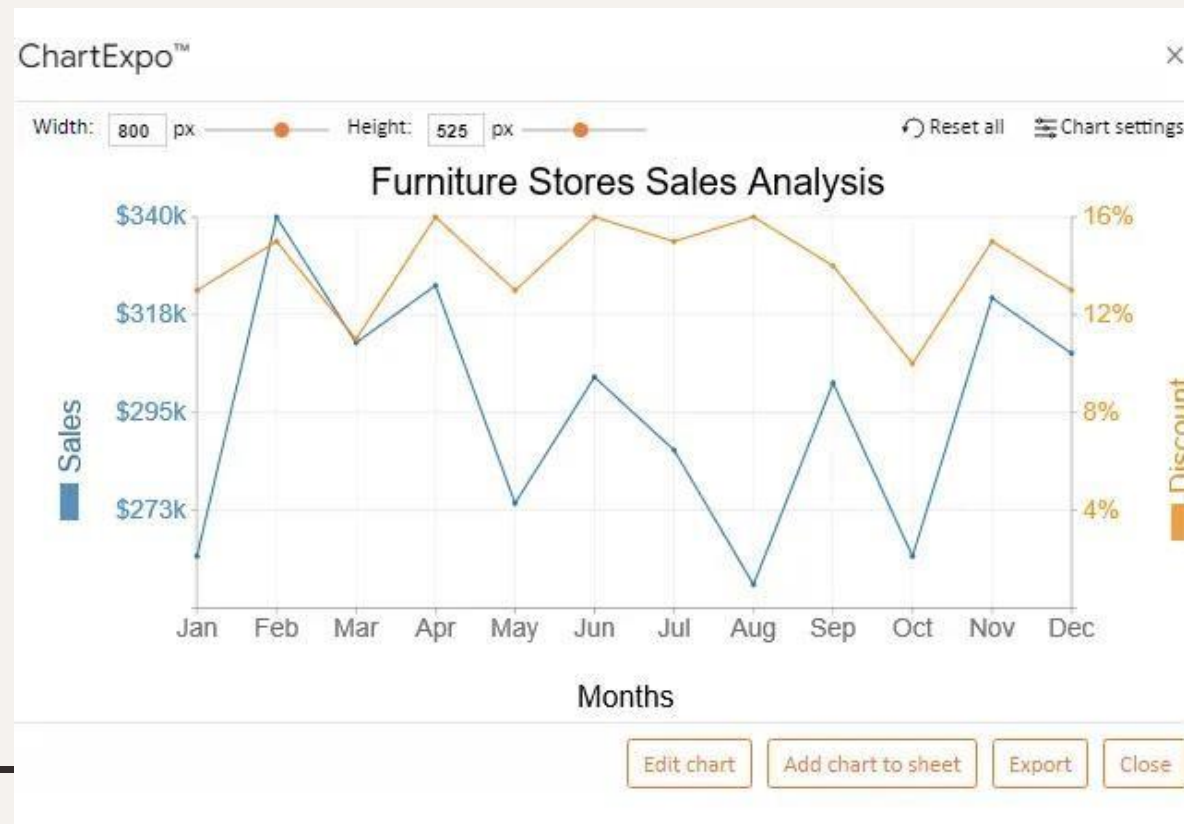
Trends: Line chart

Dates in which couples announce their break-up on Facebook



Trends: Line chart

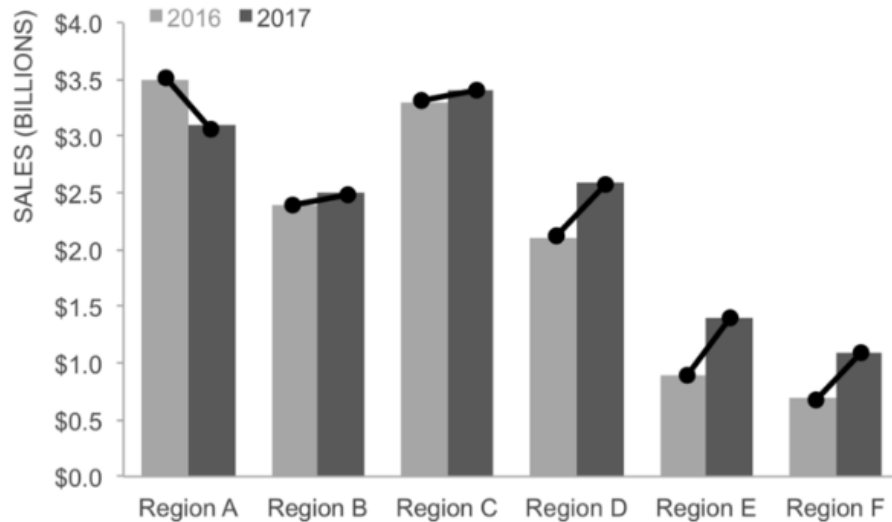
- Time is ALWAYS displayed in the X axis
- **Best practice:** the independent/cause variable should be in the X axis, while the dependent variable/effect in the Y axis



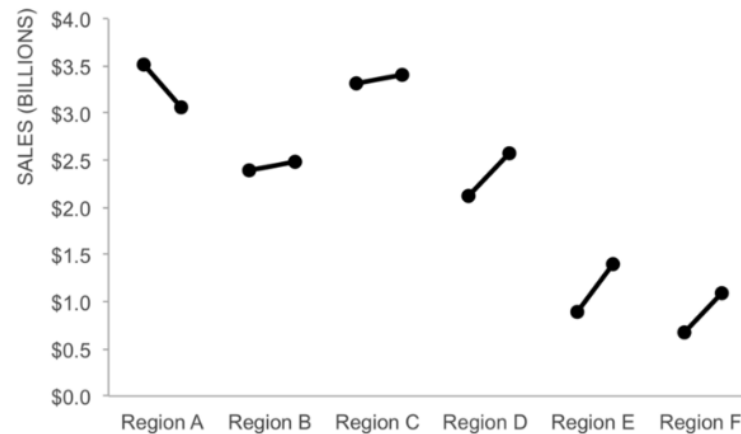
Trends: Slopegraph

- Useful to communicate overall change
- When you have two time points only
- Multiple categories to compare
- Direction of change is the story

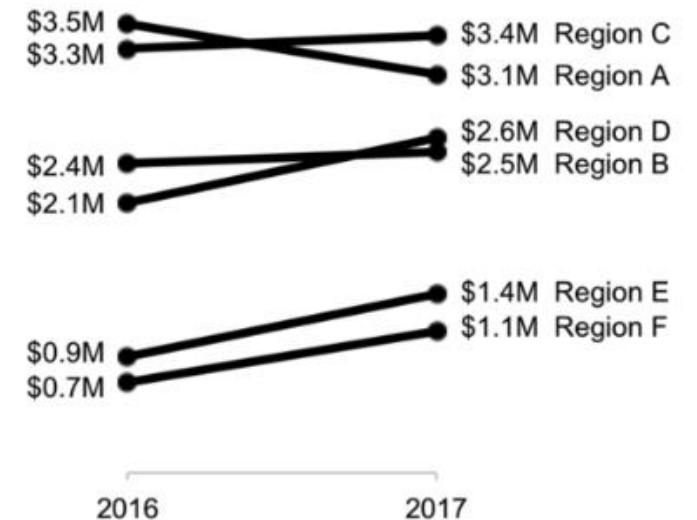
Sales by region



Sales by region

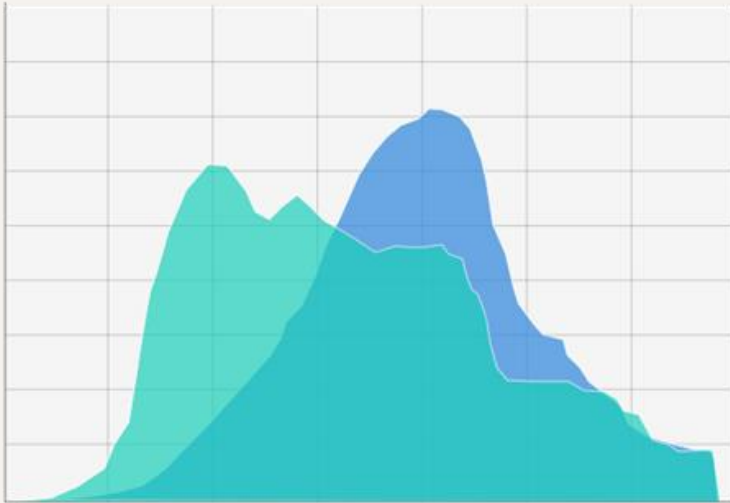


Sales by region
BILLIONS

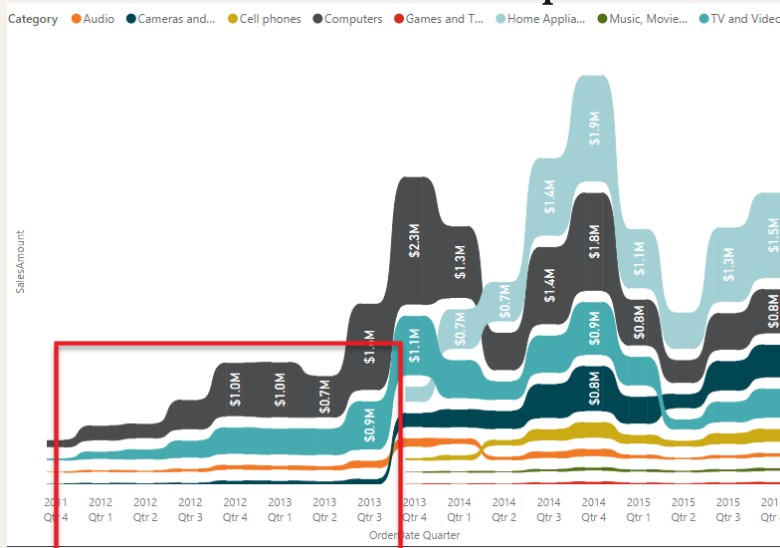


Trends: Area chart, spiral plot, ribbon charts

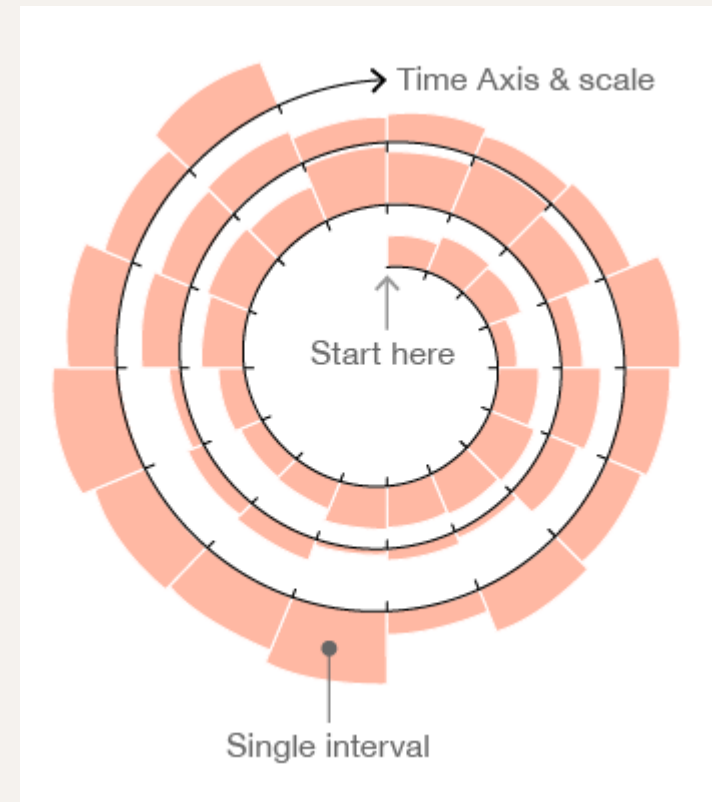
Combines a line chart and a bar chart to show changes in quantities over time.



Ribbon charts are effective at showing how rank can change. The ribbon for the highest rank (value) is always displayed on top of the other ribbons for each time period.



Spiral Plots are great for displaying periodic patterns.



Part-to-whole relationships

SPOILER:

We are better at calculating the area of rectangles, but research has shown that **humans prefer rounded edges** and that people's willingness to interact and remember data is affected by the aesthetics

When to use: You need to show how individual parts make up a total.

Best visuals:

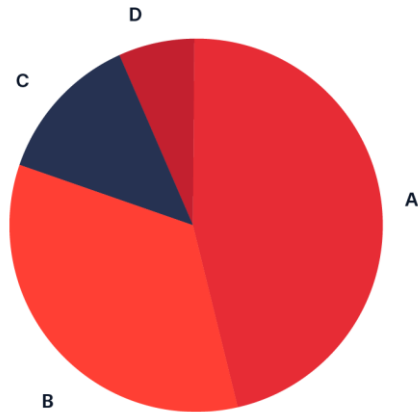
- **Pie and donut charts**: Best with fewer than eight categories. Use when the precise comparison between slices isn't critical, but the part-to-whole relationship is important.
- **Treemaps**: Display hierarchical data as nested rectangles. Excellent for showing proportions when you have many categories or need to show hierarchy.
- **100% stacked column or bar charts**: Compare proportions across multiple groups. Each bar represents 100% divided among components.



Proportions/Part-to-whole: **Pie chart**

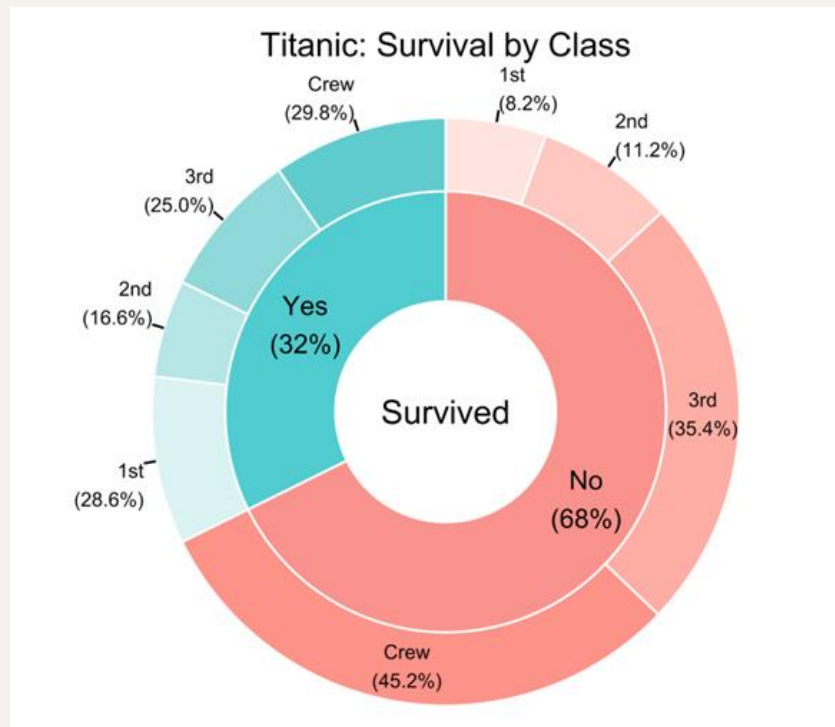
- How much C is larger than D?

When **specificity** is not that important, we can opt for a pie chart.



Avoid: Pie and donut charts when you need precise comparison. People find it difficult to compare angles and areas accurately.

Proportions/Part-to-whole: Donut chart

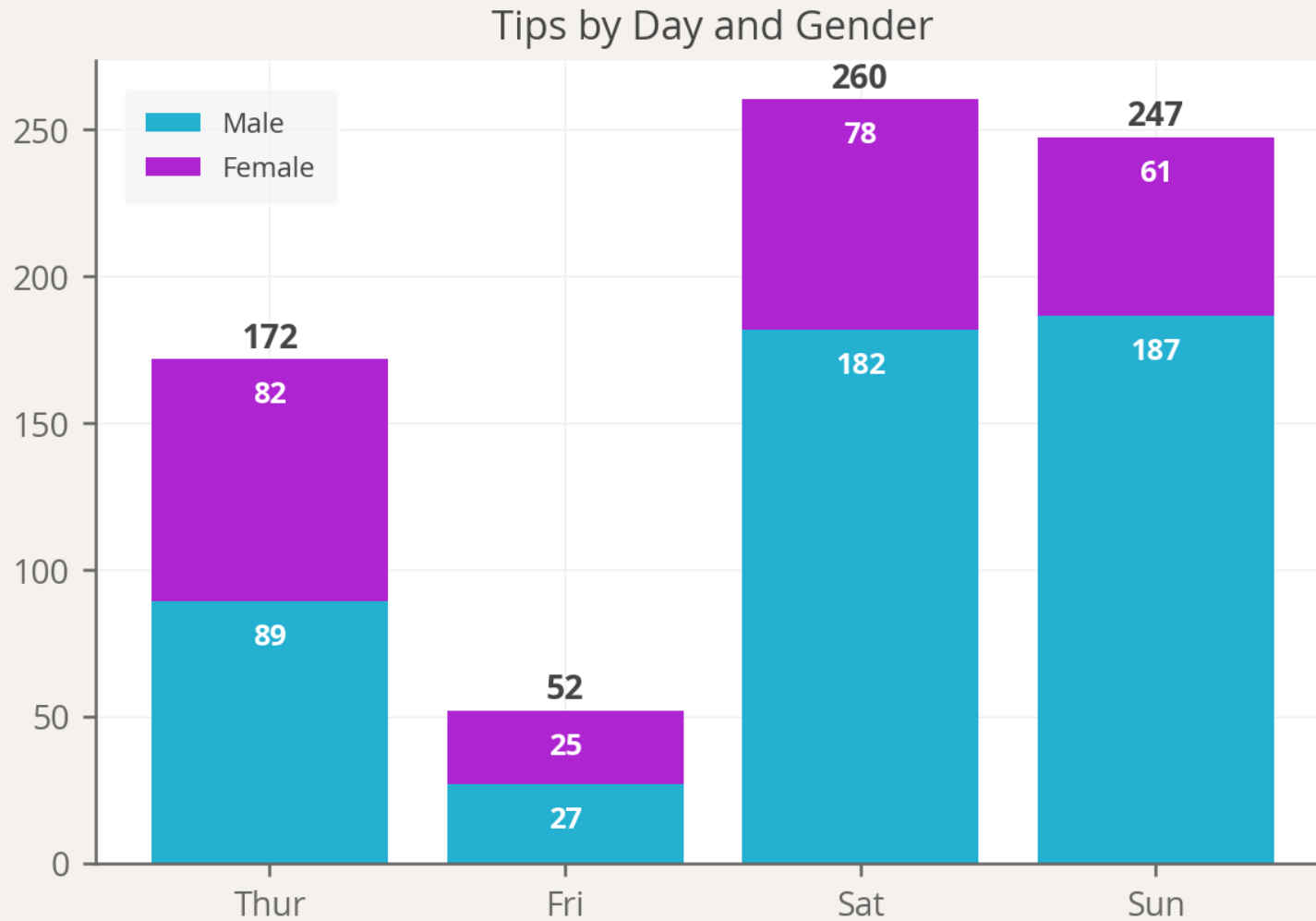


Jastrow illusion



BEWARE!

Proportions without a circular shape: **Stacked bar**

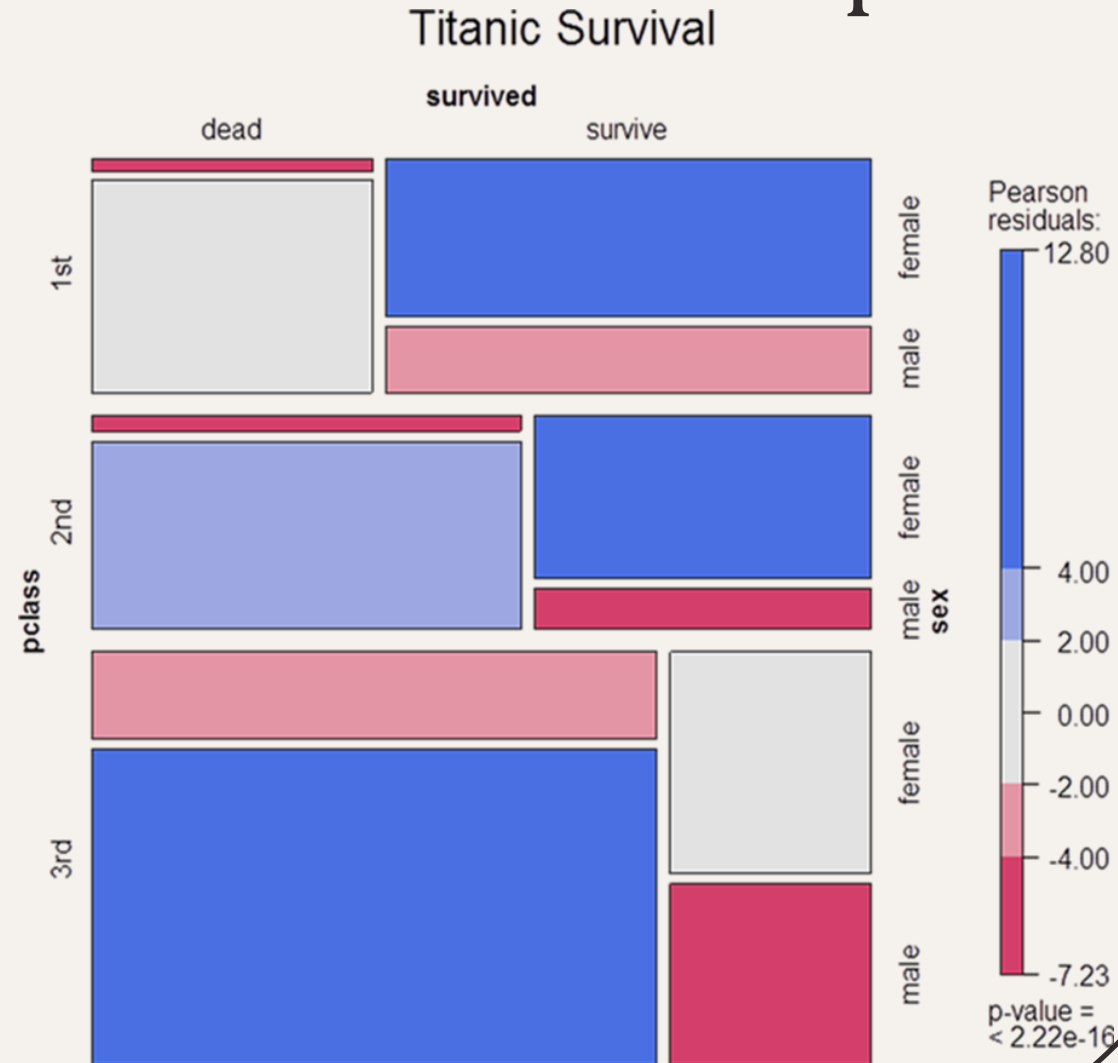


Proportions without a circular shape: Tree map

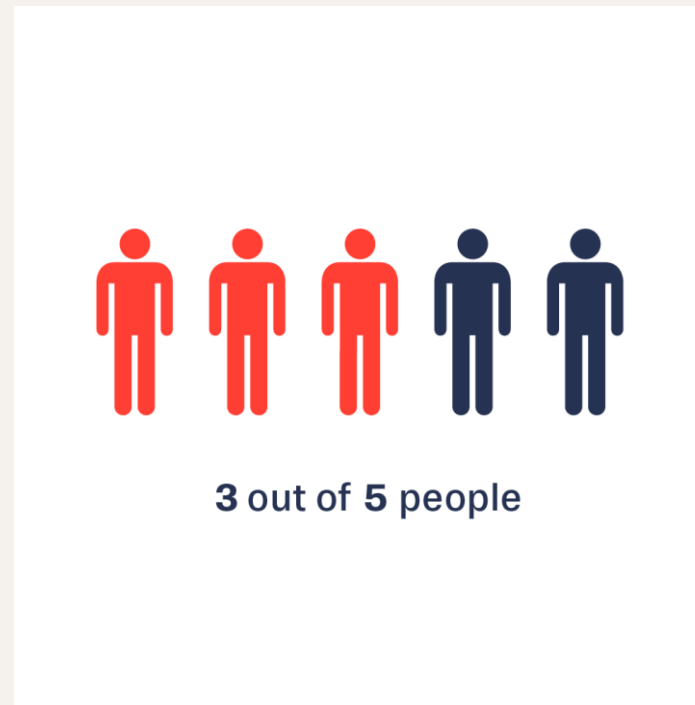
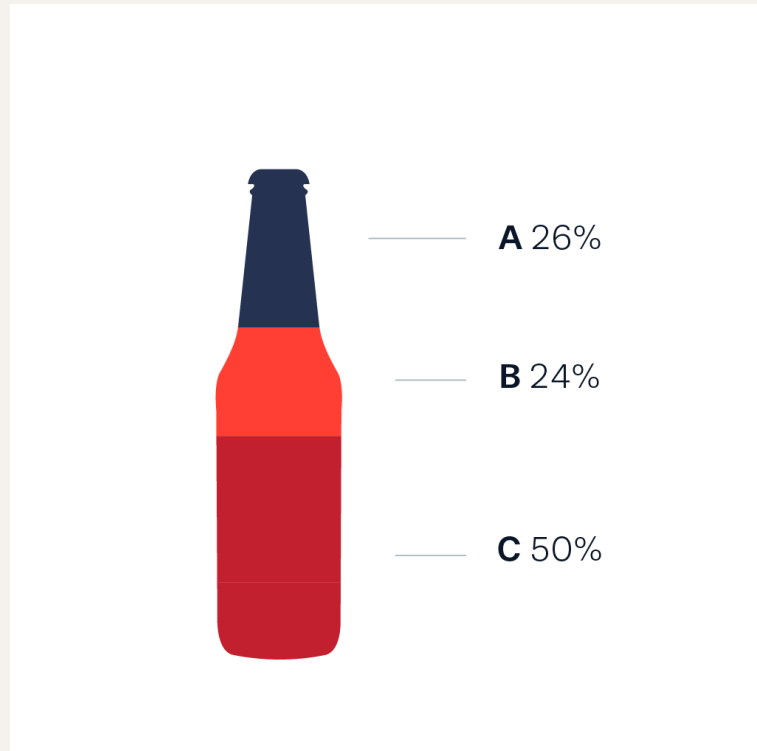
Dark blue represents more cases than expected given independence.

Dark red represents less cases than expected if independence holds.

If class, gender, and survival are independent, we are seeing many more male crew perishing, and 1st, 2nd and 3rd class females surviving than would be expected.



Proportions without a circular shape: **Pictorial stacked bar** and **pictorial fraction chart/isotypes**



Only for
whole
numbers!

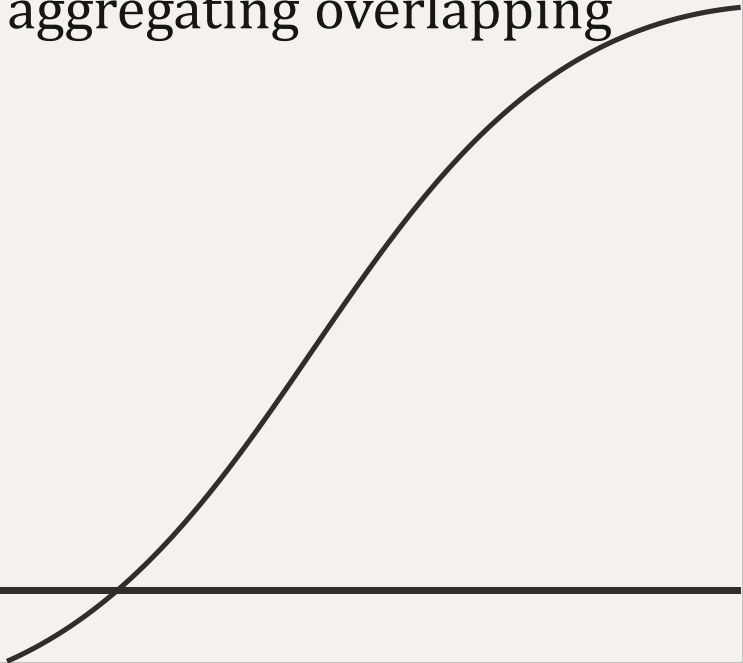
Distribution and relationships

When to use: You need to understand how data is distributed or identify relationships between variables.

Best visuals:

- **Scatter, bubble, and dot plot charts**: Show relationships between two numerical variables. Look for correlations, clusters, and outliers. Bubble charts add a third dimension through size.
- **High-density scatter charts**: Handle large datasets efficiently by aggregating overlapping data points.

Example scenarios:

- Price vs. sales volume
 - Customer satisfaction vs. purchase frequency
 - Product cost vs. profit margin
 - Distribution of customer ages
 - Analyzing large datasets with thousands of data points
- 
-

Relationships: **Correlation**

- Relationship between more than one variable and their values
- For example, you may be looking for the relationship between classroom size and school graduation rate, or how much lung capacity relates to endurance.
- But remember, **correlation does not always equal causation.**

Correlation

[kawr-uh-ley-shuh n]

noun

The degree to which two measurements vary together

Causation

[kaw-zey-shuh n]

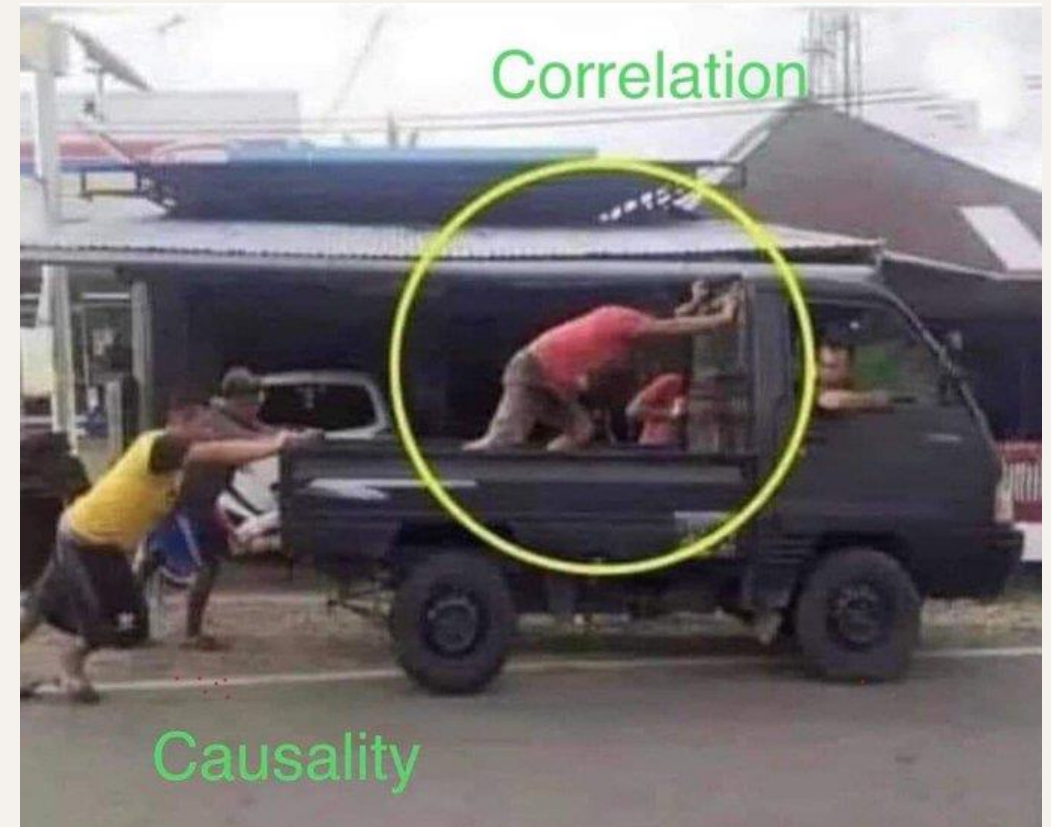
noun

The act of one thing being the result, or effect, of another thing

Correlation is not causation!



Why does the van move?



Which concept is referring this article?

Monday, June 10, 2019

Sleeping with artificial light at night associated with weight gain in women

Eliminating light while sleeping could reduce obesity.



Sleeping with a television or light on in the room may be a risk factor for gaining weight or developing obesity, according to scientists at the National Institutes of Health. The research, which was published online June 10 in JAMA Internal Medicine, is the first to find an association between any exposure to artificial light at night while sleeping and weight gain in women. The results suggest that cutting off lights at bedtime could reduce women's chances of becoming obese.

The research team used questionnaire data from 43,722 women in the Sister Study, a cohort study that examines risk factors for breast cancer and other diseases. The participants, aged 35-74 years, had no history of cancer or cardiovascular disease and were not shift workers, daytime sleepers, or pregnant when the study began. The study questionnaire asked whether the women slept with no light, a small nightlight, light outside of the room, or a light or television on in the room.

The scientists used weight, height, waist and hip circumference, and body mass index measurements taken at baseline, as well as self-reported information on weight at baseline and follow-up five years later. Using this information, the scientists were able to study obesity and weight gain in women exposed to artificial light at night with women who reported sleeping in dark rooms.

The results varied with the level of artificial light at night exposure. For example, using a small nightlight was not associated with weight gain, whereas women who slept with a light or television on were 17% more likely to have gained 5 kilograms, approximately 11 pounds, or more over the follow-up period. The association with having light coming from outside the room was more modest.

Also, the scientists wondered if not getting enough rest factored into the findings.

NEWS RELEASES

Monday, June 10, 2019

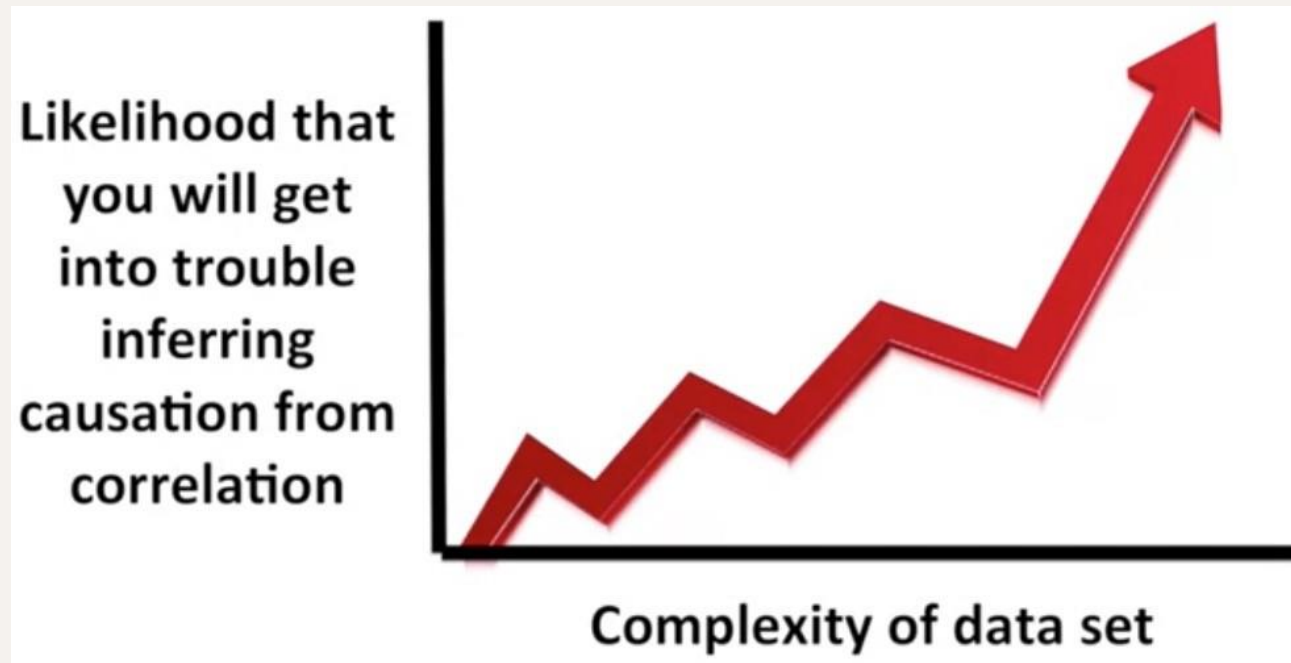
Sleeping with artificial light at night associated with weight gain in women

Eliminating light while sleeping could reduce obesity.

SOURCE: [HTTPS://WWW.NIH.GOV/NEWS-EVENTS/NEWS-RELEASES/SLEEPING-ARTIFICIAL-LIGHT-NIGHT-ASSOCIATED-WEIGHT-GAIN-WOMEN](https://www.nih.gov/news-events/news-releases/sleeping-artificial-light-night-associated-weight-gain-women)

TV LIGHT WHILE SLEEPING CAUSES OBESITY

Something to remember

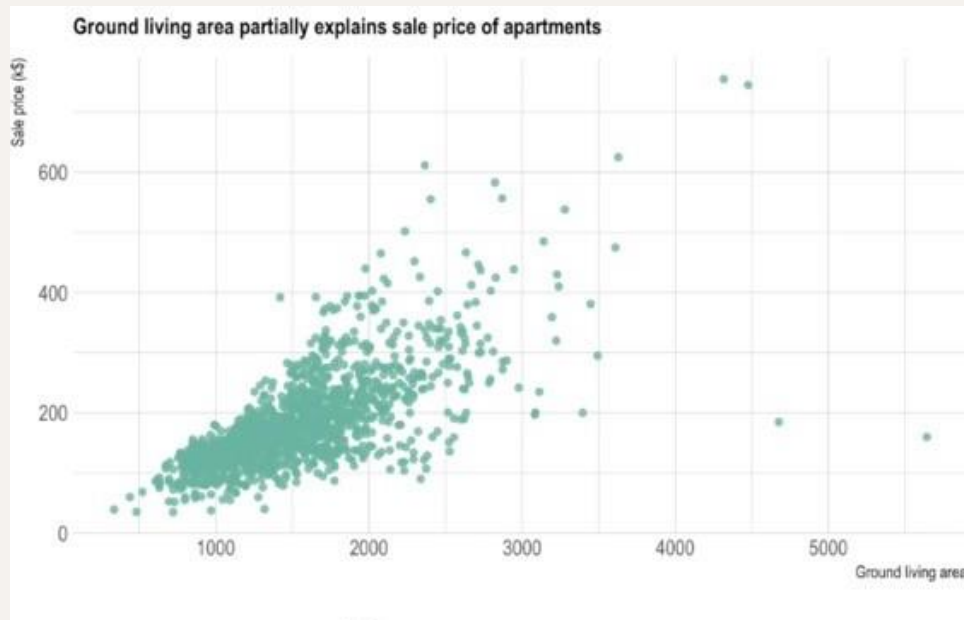


Relationships: **Correlation** - Scatterplot

- Relationship between more than one variable and their values
- **Scatterplot**: one of the most effective visualization to show the correlation between two variables

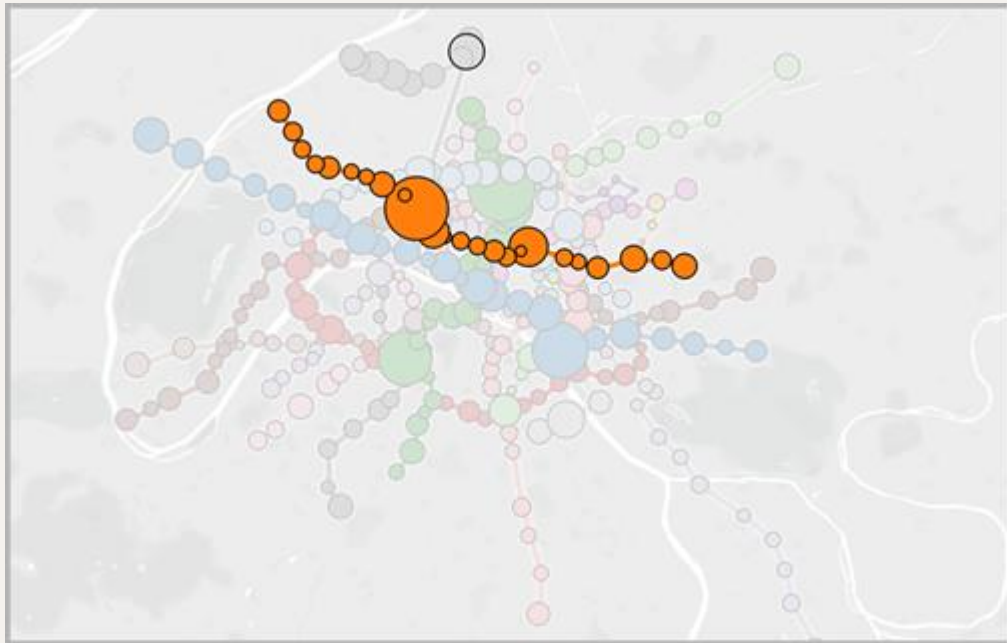
What types of question can this chart answer?

- Are these two measures related? How strongly?
- Are some measures more related than others?
- How strongly related are these measures?



Relationships: **Flow**

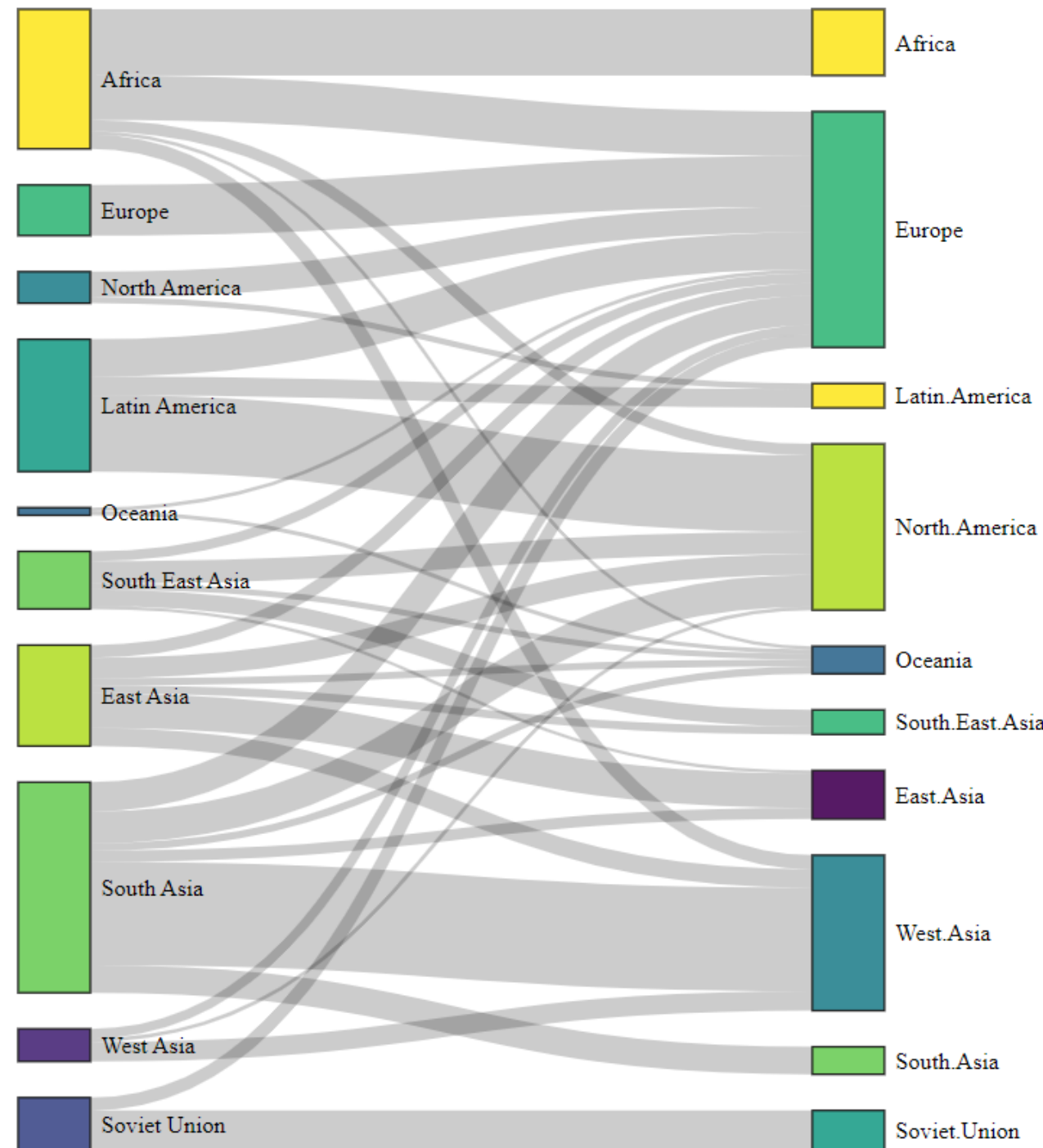
- Flow charts can be maps that convey movement over time, such as Sankey diagrams.
- Flow maps include path over time and path between origin and destination charts.



- What is the longest shipping route?
- How long are people lingering around gates?
- What are the bottlenecks to traffic in the city?

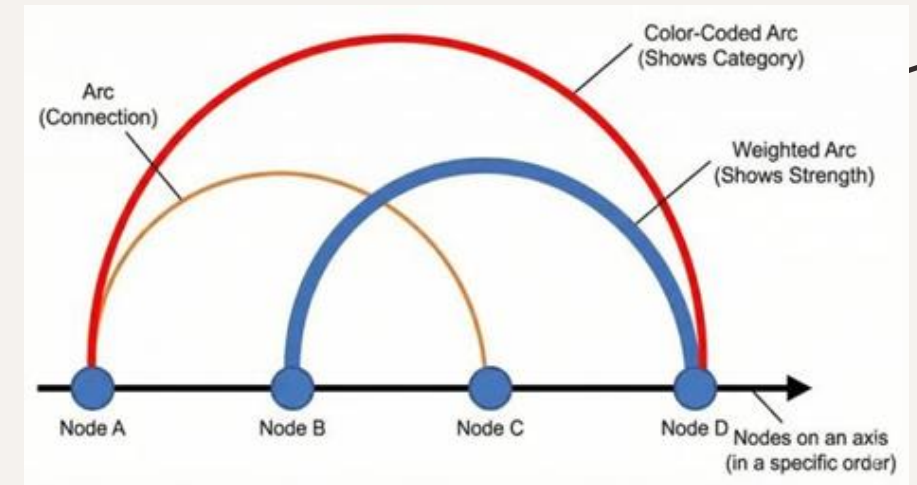
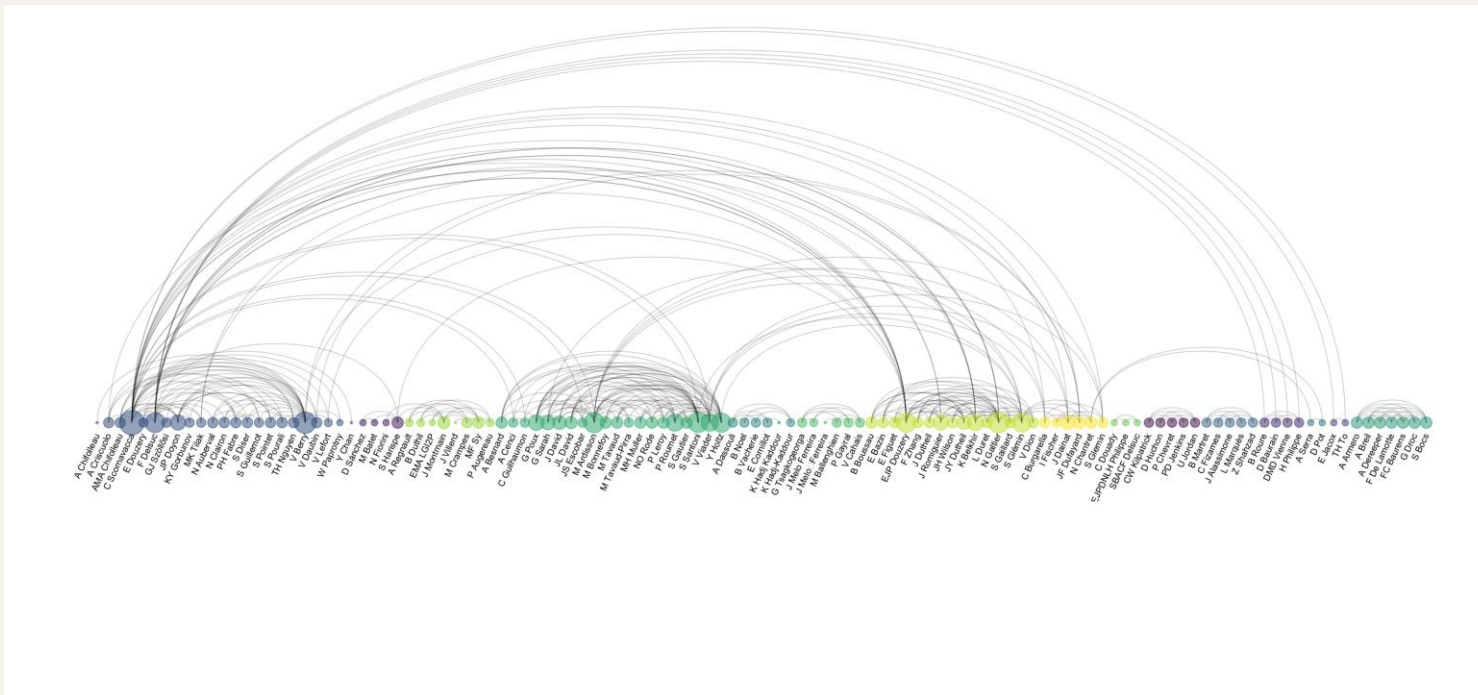
Relationships: Flow – Sankey diagram

- Several entities (nodes) are represented by rectangles or text. Their links are represented with arrow or arcs that have a width proportional to the importance of the flow
- **Example:** number of people migrating from one country (left) to another (right)
- You can show the flow and the proportional size of the flow



Relationships: Flow – Arc diagrams

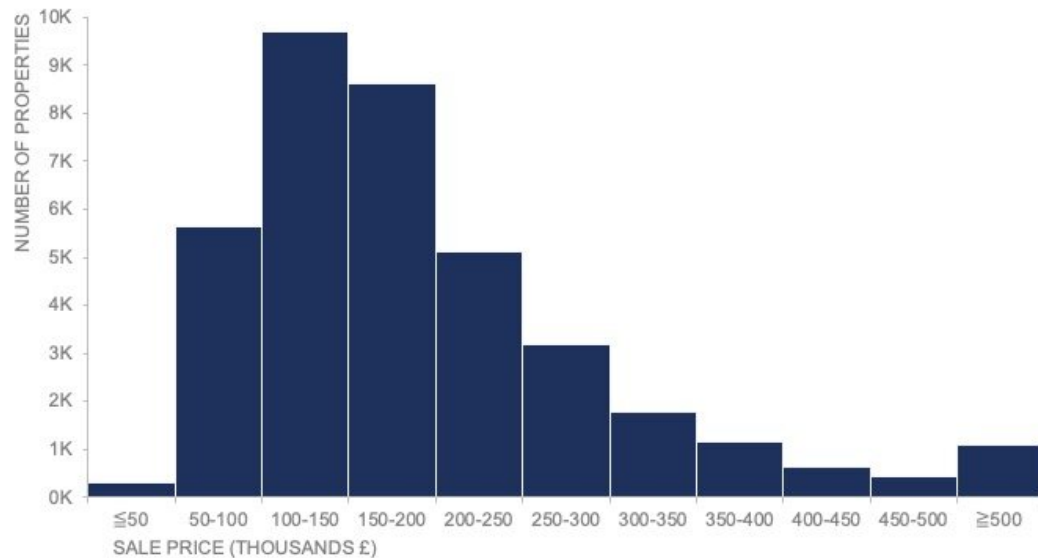
It is constituted by nodes that represent entities and by links that show relationships between entities. In arc diagrams, nodes are displayed along a single axis and links are represented with arcs.



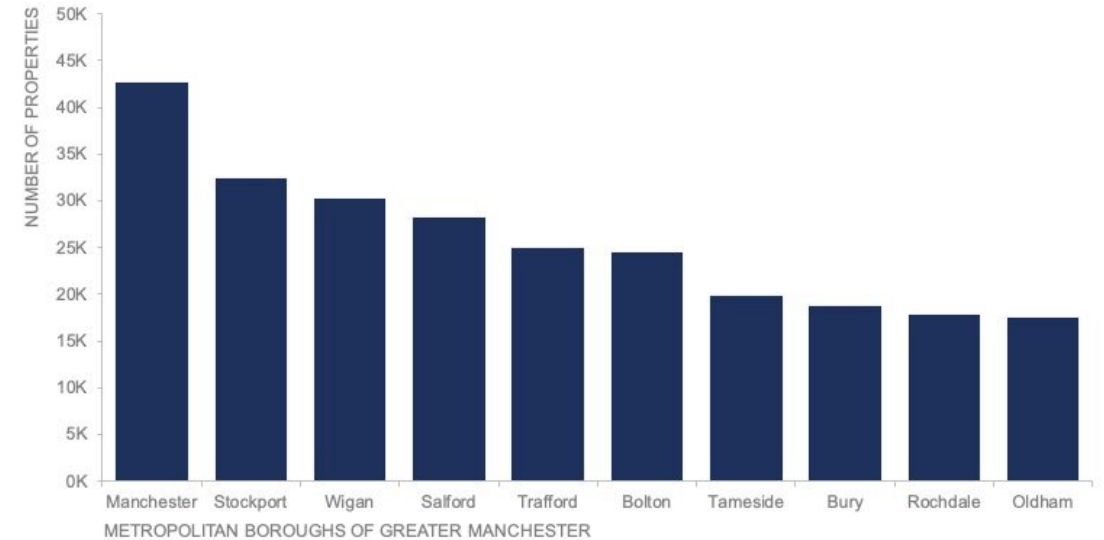
Distribution - Histogram

- Distribution vs. Comparison?
- **Histogram:** graph that shows the frequency of numerical data using rectangles. The height of a rectangle (the vertical axis) represents the distribution frequency of a variable (the amount, or how often that variable appears).

Distribution of property sales: January 2013 to September 2019

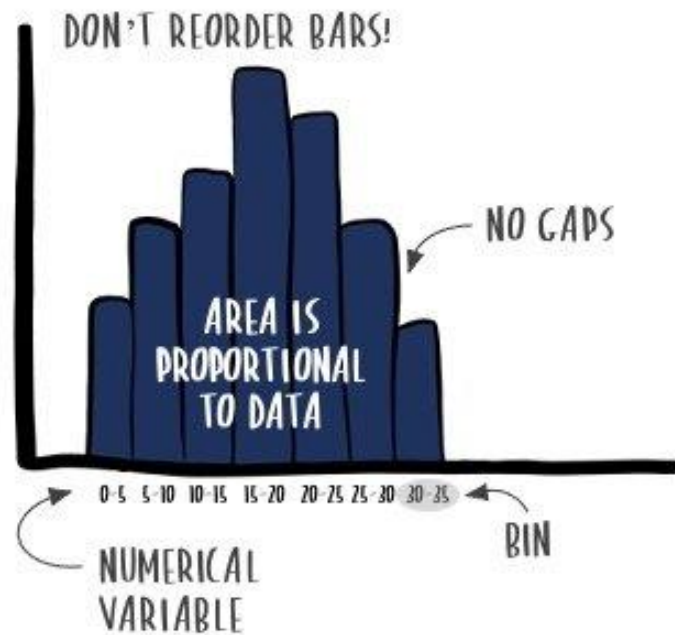


Residential property sales by location
January 2013 to September 2019

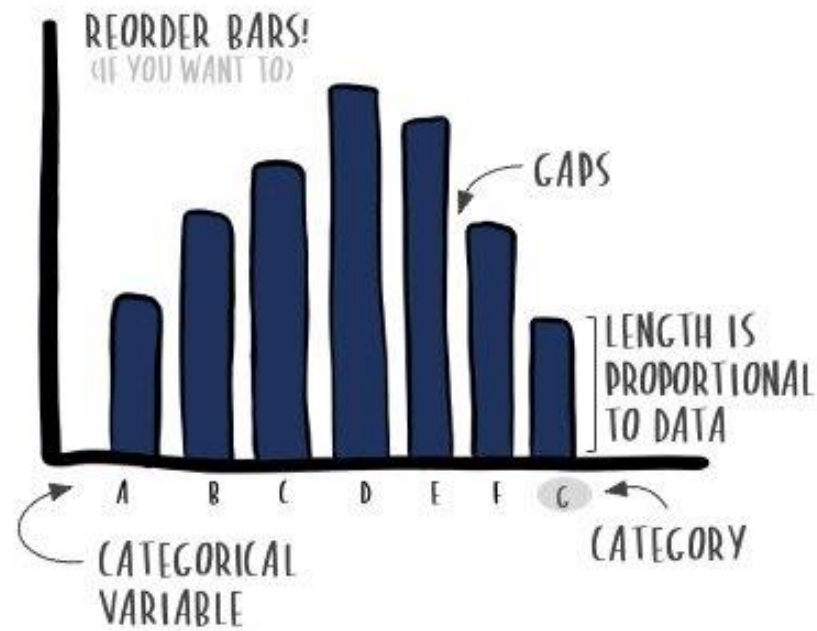


Distribution – Histogram vs. Bar chart

This is a histogram...

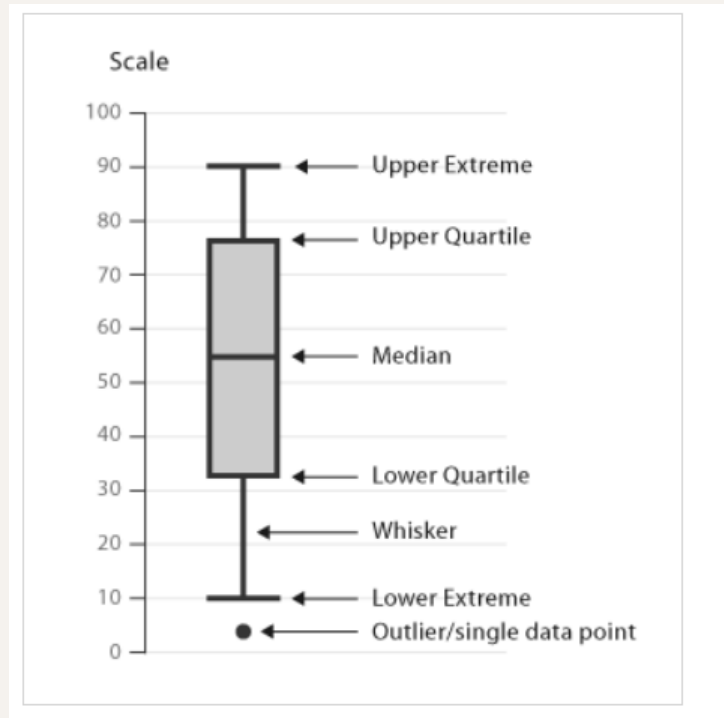


This is a bar chart...



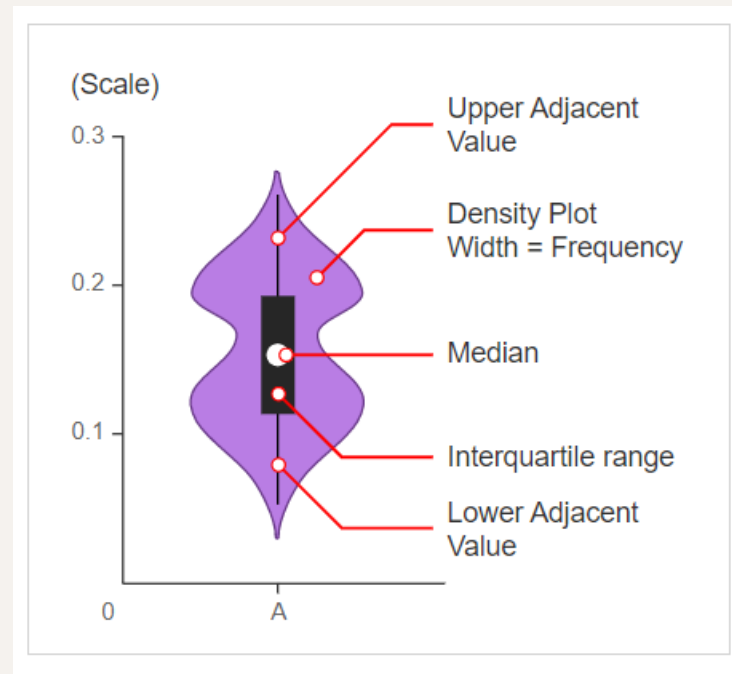
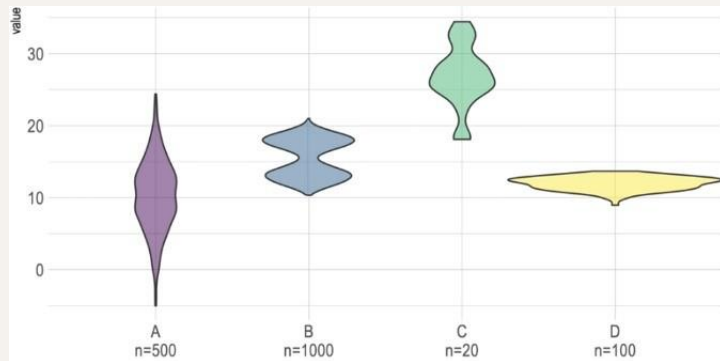
Distribution – Boxplot or Box and Whiskers plot

- A boxplot summarizes the distribution of a numeric variable through their quartiles.
- Advantage: Taking up less space, which is useful when comparing distributions between many groups or datasets.
- It emphasizes the data spread



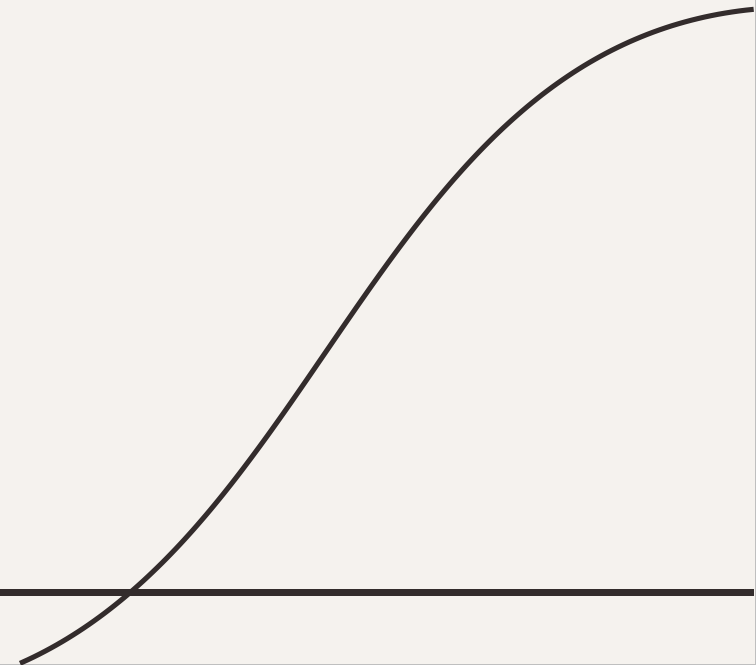
Distribution – Violin plot

- Combination of a box plot and a density plot. It is used to visualize the distribution of the data and its probability density.

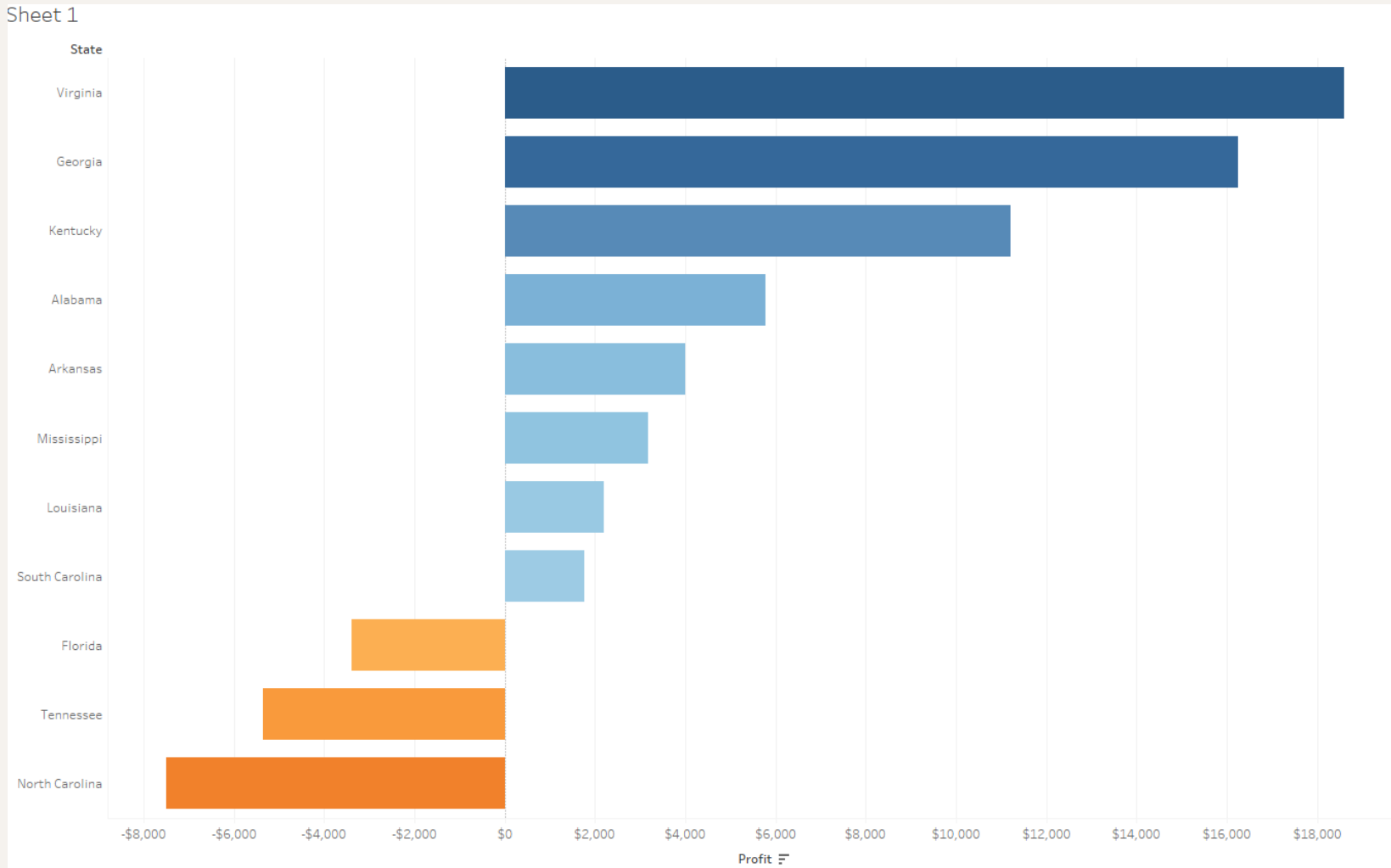


Deviation

- Deviation charts show how far a value varies from some baseline, such as the average or median.
- If you wanted to know which items had unusually high or low profit margins, you would use a deviation chart.
- You can use **bullet charts, bar charts, and combination charts** to show deviation.



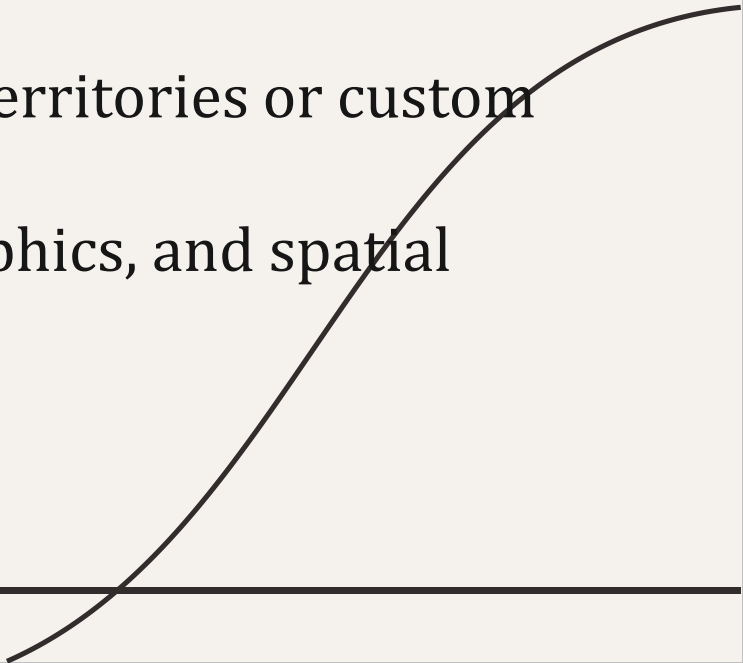
Deviation: Diverging bar chart



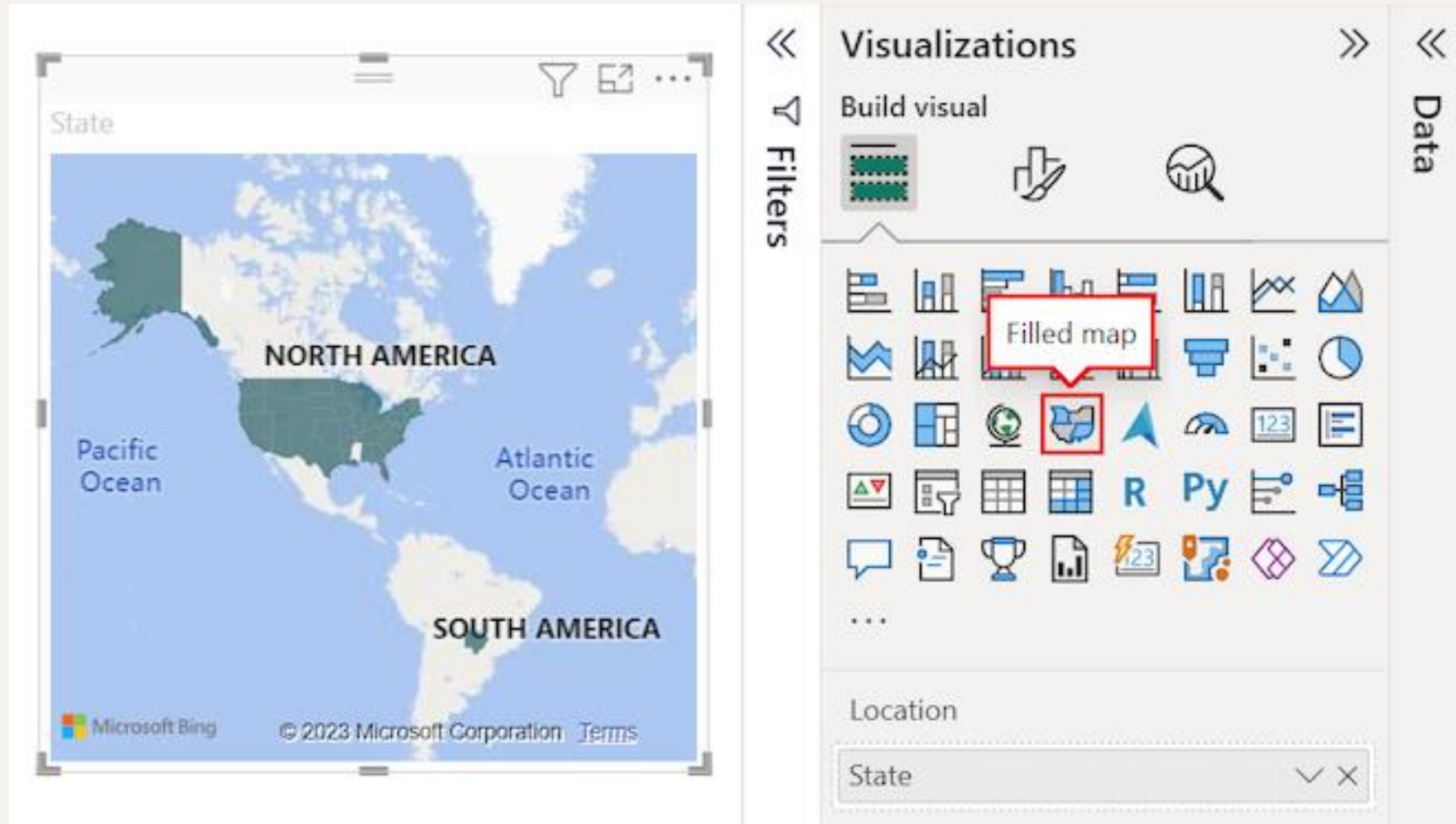
Geographic data

When to use: You have location-based data like countries/regions, states, cities, or coordinates.

Best visuals:

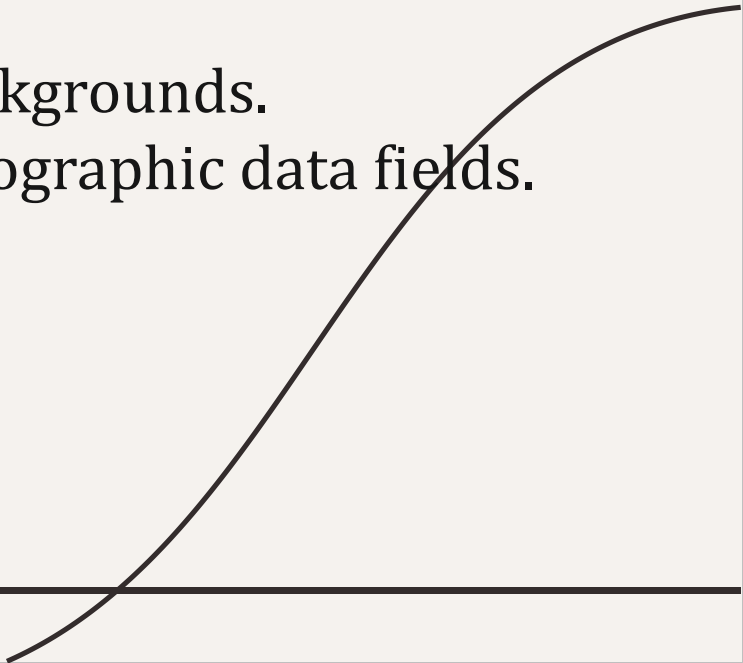
- **Filled maps**: Show data values across geographic regions using color intensity.
 - **Azure Maps**: Modern mapping with bubble, heat map, bar chart, and pie chart layers for rich geographic visualizations.
 - **Shape maps**: Use custom geographic boundaries like sales territories or custom regions.
 - **ArcGIS maps**: Advanced mapping with geocoding, demographics, and spatial analysis capabilities.
- 
-

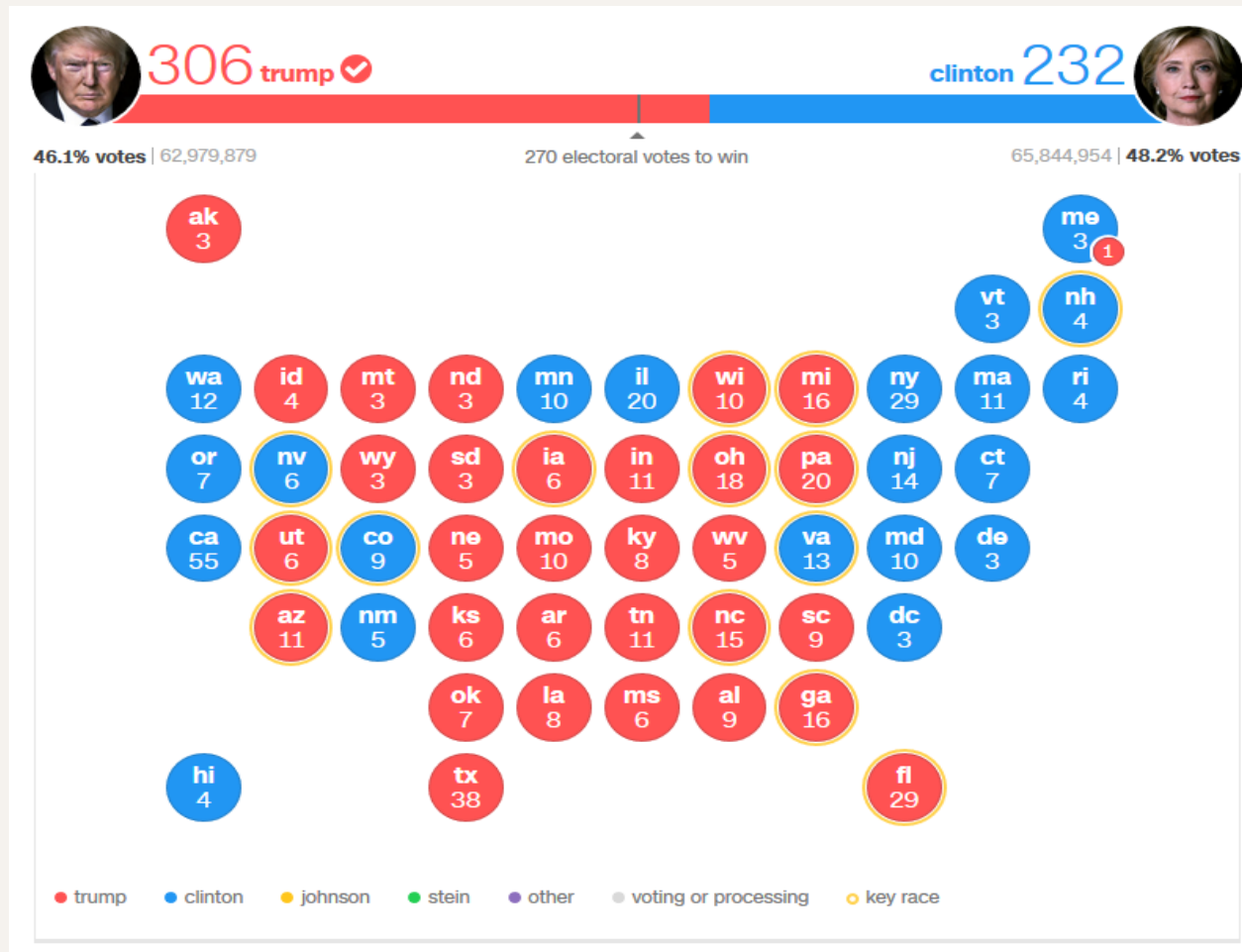
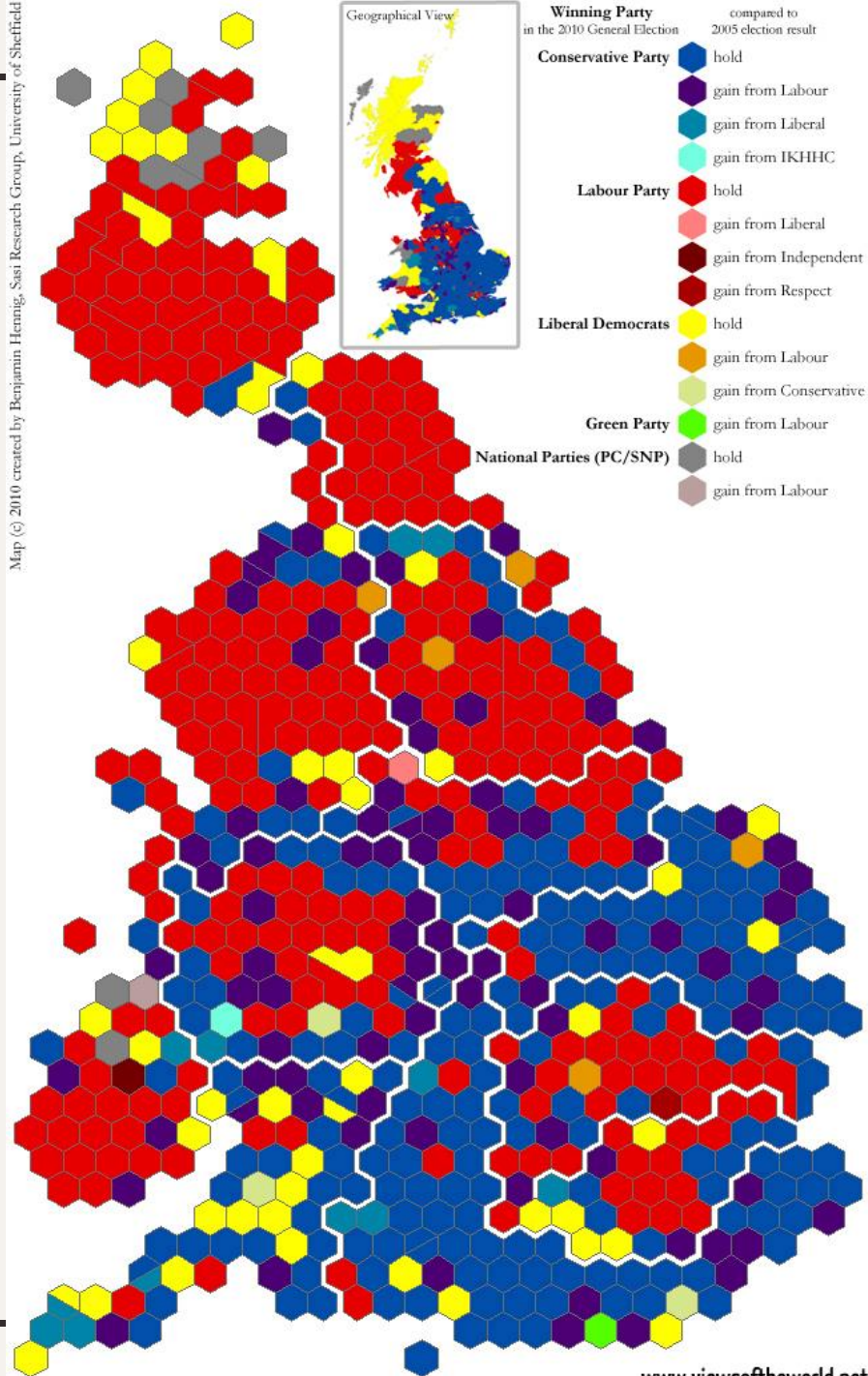
Geographic: Filled map (choropleth map)

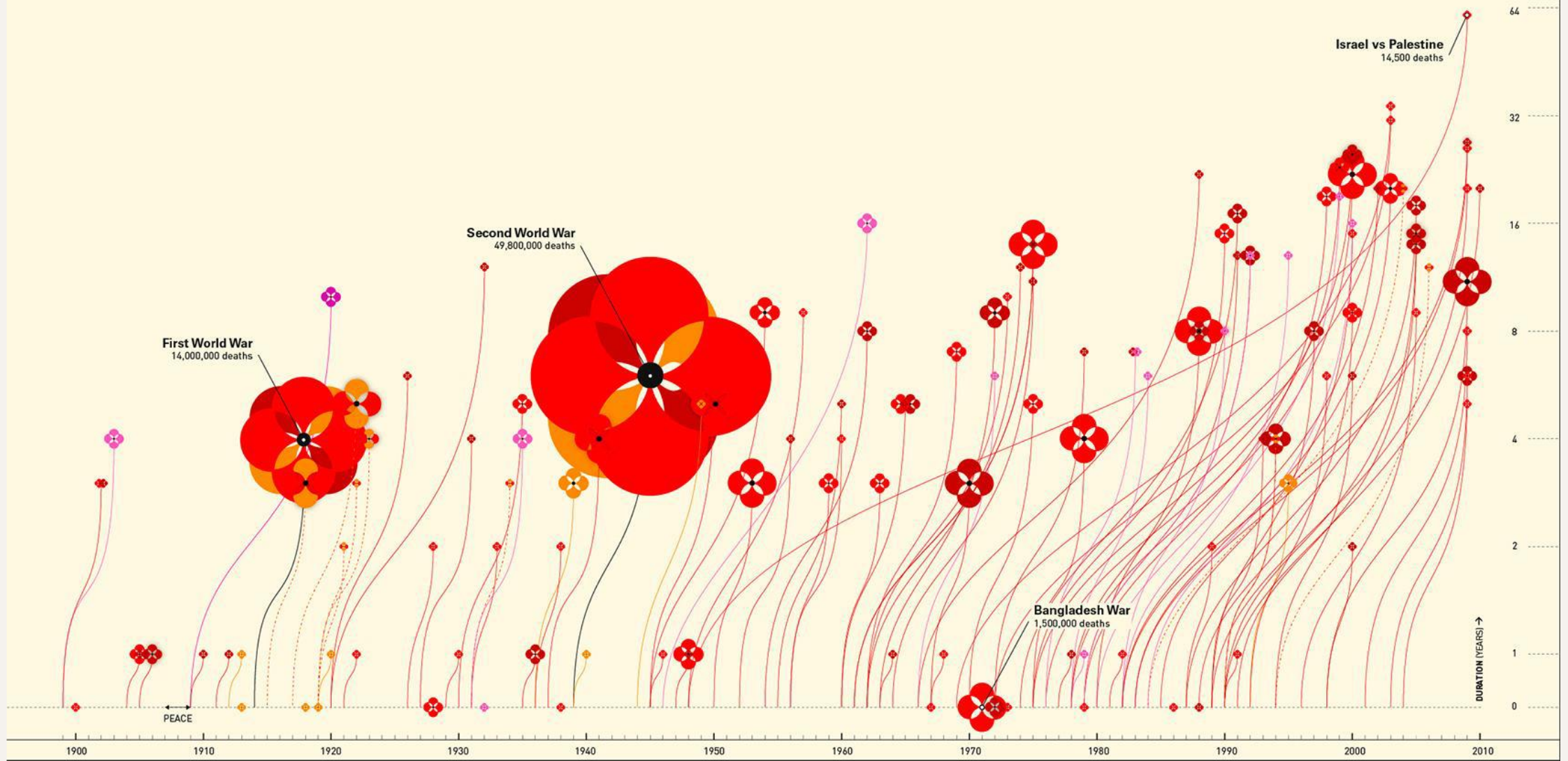


Common features across map types

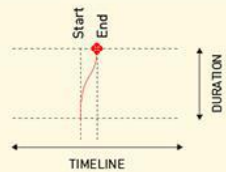
All Power BI map visualizations share common capabilities:

- **Interactivity:** Maps support cross-filtering and cross-highlighting with other visuals on your report page.
 - **Tooltips:** Hover over locations to see detailed information.
 - **Zoom and pan:** Navigate to specific areas of interest.
 - **Formatting options:** Customize colors, labels, titles, and backgrounds.
 - **Data categorization:** Power BI automatically categorizes geographic data fields.
- 





POPPY DIAGRAM



The remembrance poppy commemorates soldiers who have died in war. Each poppy in the diagram depicts a war of the last century (with more than 10,000 deaths). The stem grows from the year when the war started. The poppy flowers in the year the war ended. Its size shows the number of deaths.

NUMBER OF DEATHS IN THOUSANDS (POPPY'S SIZE)



REGIONS INVOLVED IN WARS (POPPY'S COLOUR)



A summary on visuals in Power BI:

Category	When to use
<u>Categorical comparisons</u>	Show how values differ across categories or segments.
<u>Time-based trends</u>	Reveal patterns and changes over time.
<u>Part-to-whole relationships</u>	Show how components contribute to a total.
<u>Distribution and relationships</u>	Understand how data is distributed or identify relationships between variables.
<u>Sequential processes</u>	Show stages in a process where items flow from one stage to the next.
<u>Geographic data</u>	Present location-based data with geographic or spatial patterns.
<u>Key metrics and progress</u>	Highlight a single important value or track progress toward a goal.
<u>Detailed data and exact values</u>	Show exact numbers, multiple measures, or enable detailed record drill-down.
<u>AI-powered exploration</u>	Identify patterns, anomalies, or influencers with AI assistance.
<u>Slicers and filtering</u>	Provide on-canvas filtering controls for interactive data exploration.
<u>Other specialized visuals</u>	Use specialized functionality beyond standard data visualization.